

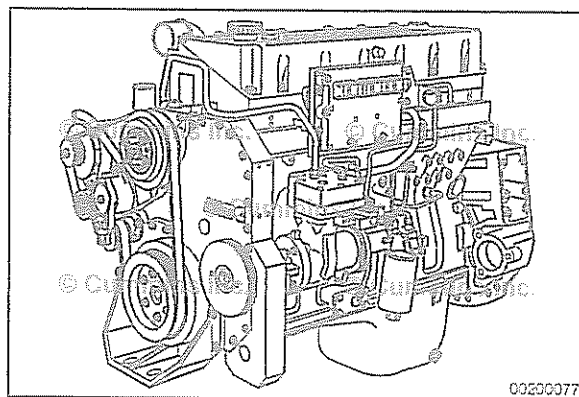
Engine Identification

Engine Dataplate

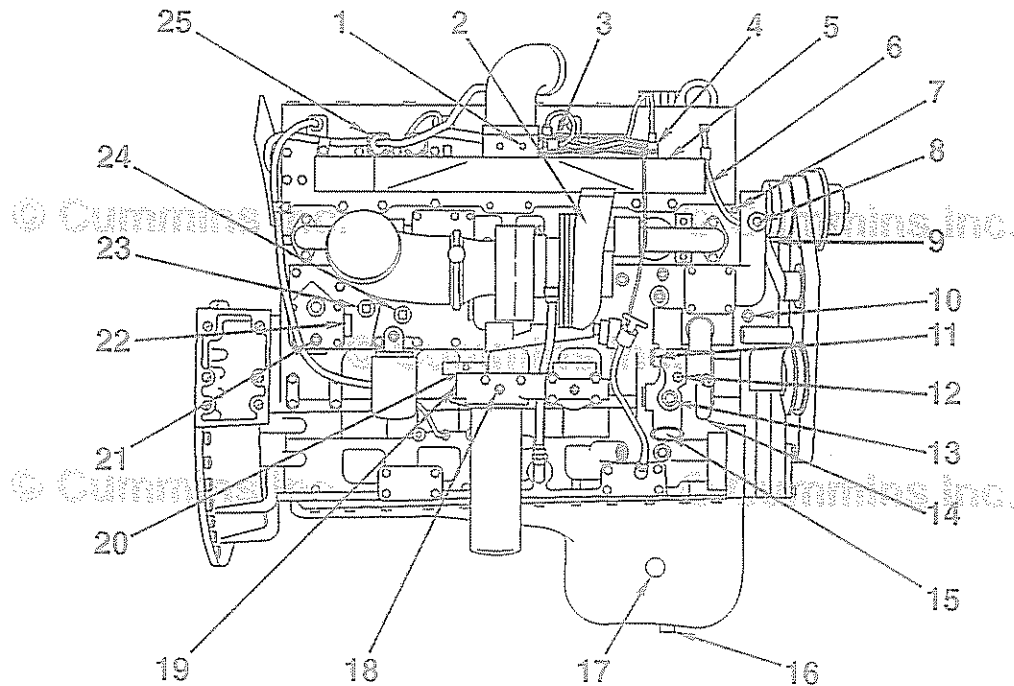
Marine Applications

The engine dataplate shows important facts about your engine. The engine serial number and CPL provide data for ordering parts and service. The engine dataplate must not be changed unless approved by Cummins Inc.

The dataplate is located on the fuel pump side of the engine, on the rocker housing. Have the following engine data available when communicating with a Cummins Authorized Repair Location. The following information on the dataplate is mandatory when sourcing service parts:



Industrial and Generator Drive

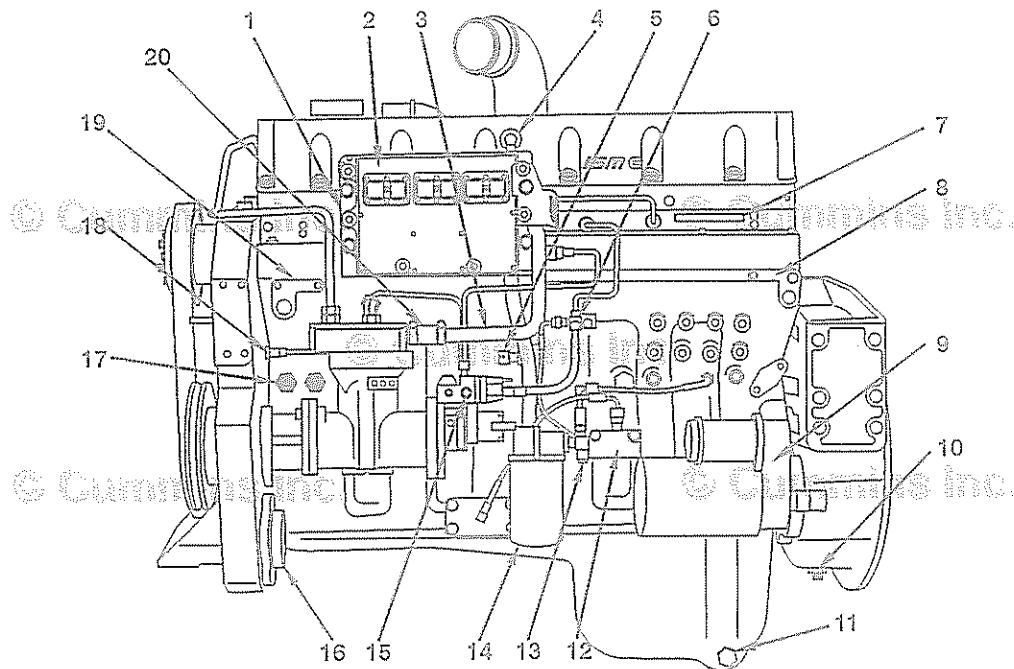


Industrial and Generator Drive - Exhaust Side

- | | |
|--|--------------------------------|
| 1. Ether hookup | 14. Water pump pressure |
| 2. Before charge air cooler | 15. Coolant inlet |
| 3. Boost temperature sensor | 16. Oil drain |
| 4. Intake manifold pressure sensor | 17. Oil pan pump heater |
| 5. After charge air cooler (intake manifold) | 18. Filter outlet pressure |
| 6. Shutter stat | 19. Engine brake oil supply |
| 7. Engine vent coolant | 20. Filter inlet pressure |
| 8. Shutter stat | 21. Block coolant pressure |
| 9. Fan sensor | 22. Coolant heater |
| 10. Coolant temperature sensor | 23. Heater supply |
| 11. Heater return | 24. Coolant temperature pickup |
| 12. Coolant inlet pressure | 25. Wastegate controller. |
| 13. Makeup line from radiator | |

01200109

Industrial and Generator Drive

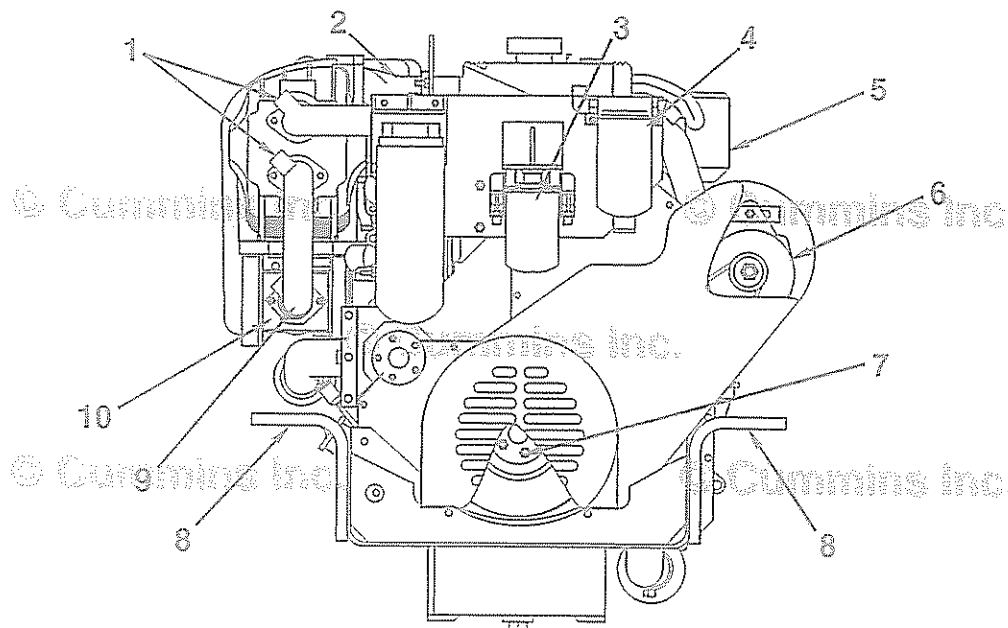


Q1200141

Industrial and Generator Drive - Fuel Pump Side

- | | |
|-------------------------------------|--|
| 1. ECM cooling plate | 11. Side oil drain |
| 2. ECM | 12. Centinel™ (optional) |
| 3. Compressor air inlet | 13. Fuel inlet to pump |
| 4. Blowby measurement | 14. Fuel filter |
| 5. Ambient air pressure sensor | 15. Rail pressure |
| 6. Fuel return to tank | 16. Power steering pump mounting location |
| 7. Engine data tag | 17. Oil pressure and temperature sensor |
| 8. Engine serial number | 18. Engine position sensor |
| 9. Starter | 19. Freon compressor mounting location. |
| 10. Flywheel ring gear speed sensor | 20. Compressor air discharge (not visible) |

Marine Applications

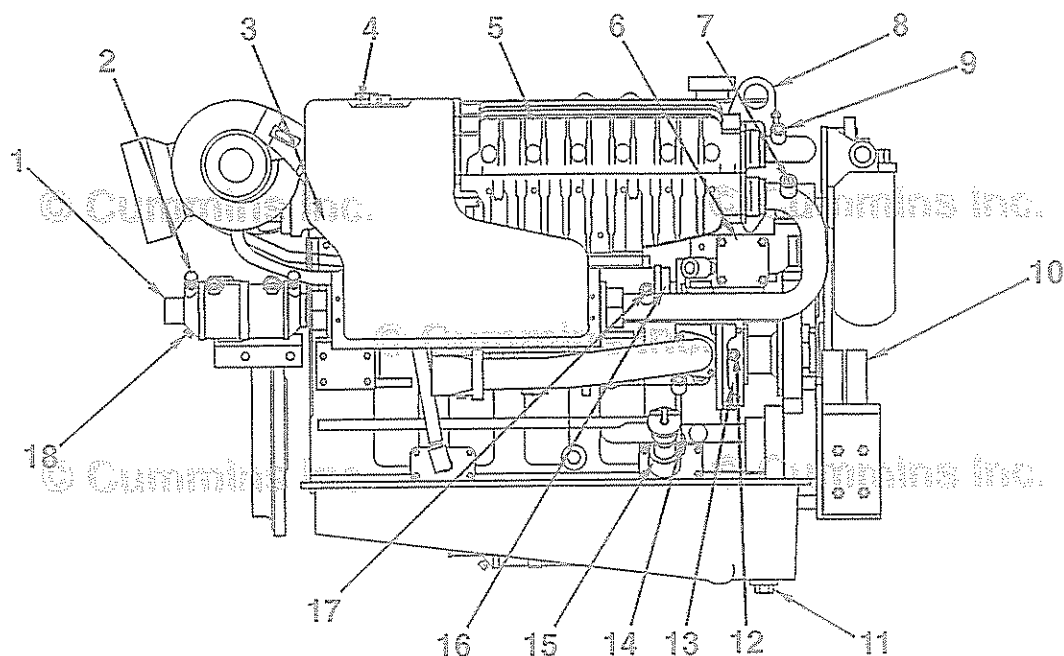


Marine - Front View

17200015

- | | |
|-----------------------|---------------------------------------|
| 1. Zinc plugs | 6. Alternator |
| 2. Air crossover | 7. Vibration damper |
| 3. Coolant filter | 8. Front engine mounts |
| 4. Fuel Filter | 9. Aftercooler-to-heat-exchanger pipe |
| 5. Crankcase breather | 10. Heat exchanger. |

Marine Applications

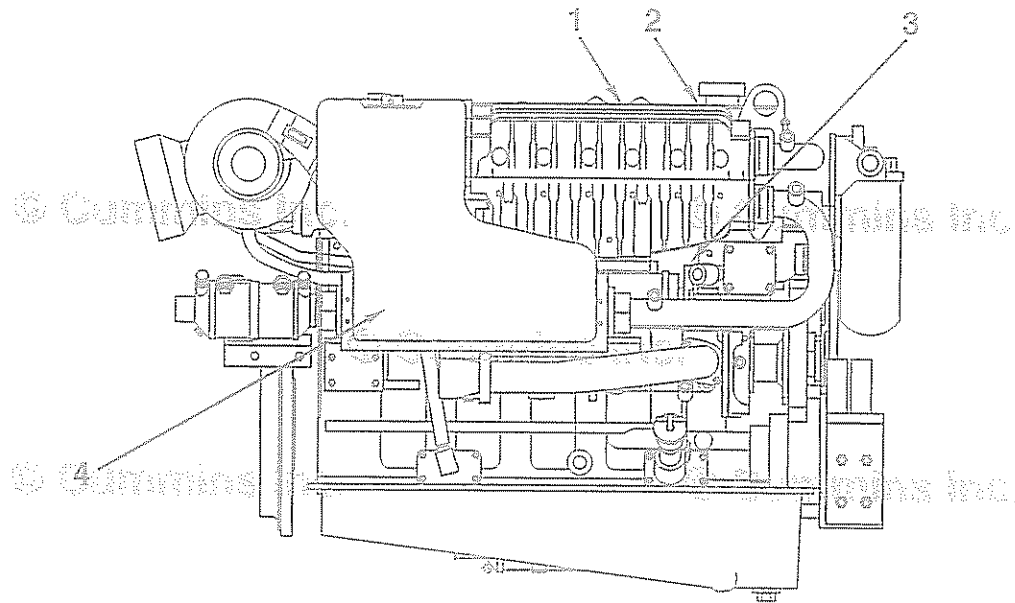


17200014

Marine - Starboard View

- | | |
|-----------------------------|---------------------------------|
| 1. Gear oil cooler | 10. Vibration damper |
| 2. Zinc anode (gear cooler) | 11. Oil pan drain plug |
| 3. Heat exchanger | 12. Coolant drain |
| 4. Expansion tank | 13. Water pump |
| 5. Aftercooler | 14. Oil dipstick |
| 6. Thermostat housing | 15. Oil fill |
| 7. Zinc anode | 16. Engine oil cooler |
| 8. Engine lifting bracket | 17. Zinc anode |
| 9. Zinc anode | 18. Gear oil cooler drain plug. |

Marine Applications

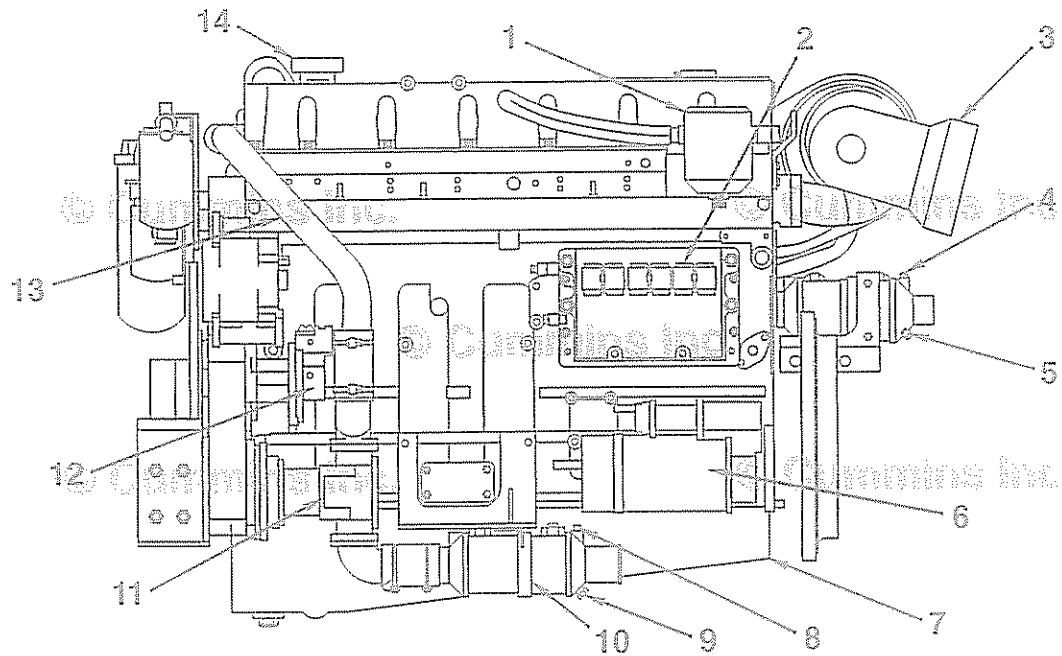


QSM11 Starboard View

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1. Intake manifold temperature sensor
2. Intake manifold pressure sensor
3. Coolant temperature sensor
4. Coolant level sensor

Marine Applications

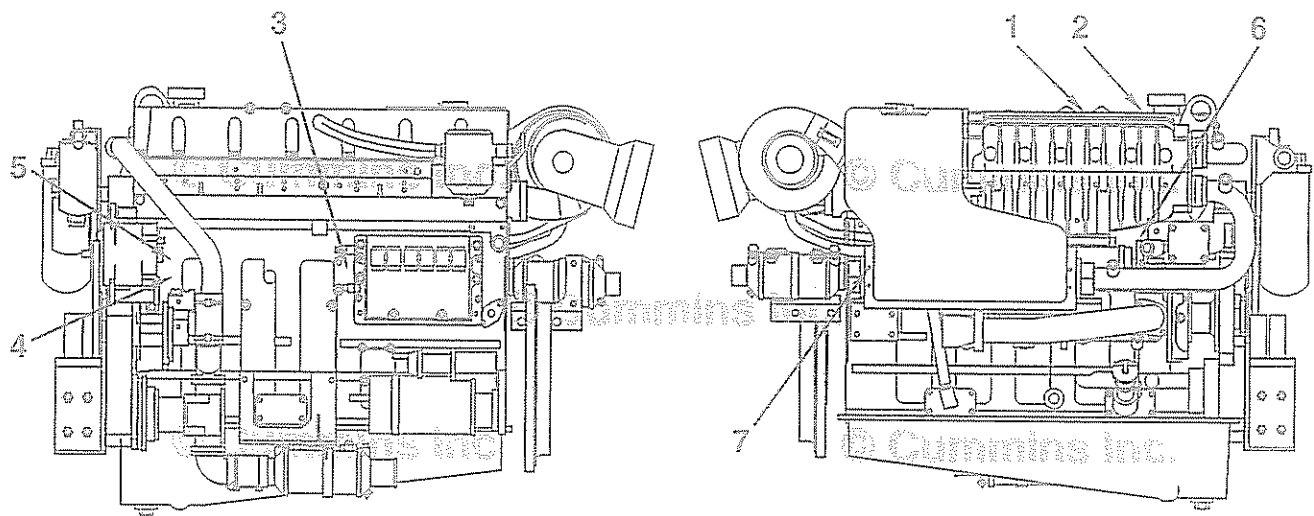


17200015

Marine - Port View

- | | |
|---------------------------|--|
| 1. Crankcase breather | 8. Zinc anode (fuel cooler) |
| 2. ECM | 9. Fuel cooler drain plug |
| 3. Exhaust elbow | 10. Fuel cooler |
| 4. Zinc anode | 11. Sea water pump |
| 5. Gear cooler drain plug | 12. Fuel pump |
| 6. Starter | 13. Sea-water-pump-to-aftercooler pipe |
| 7. Oil pan | 14. Oil fill. |

Marine Applications

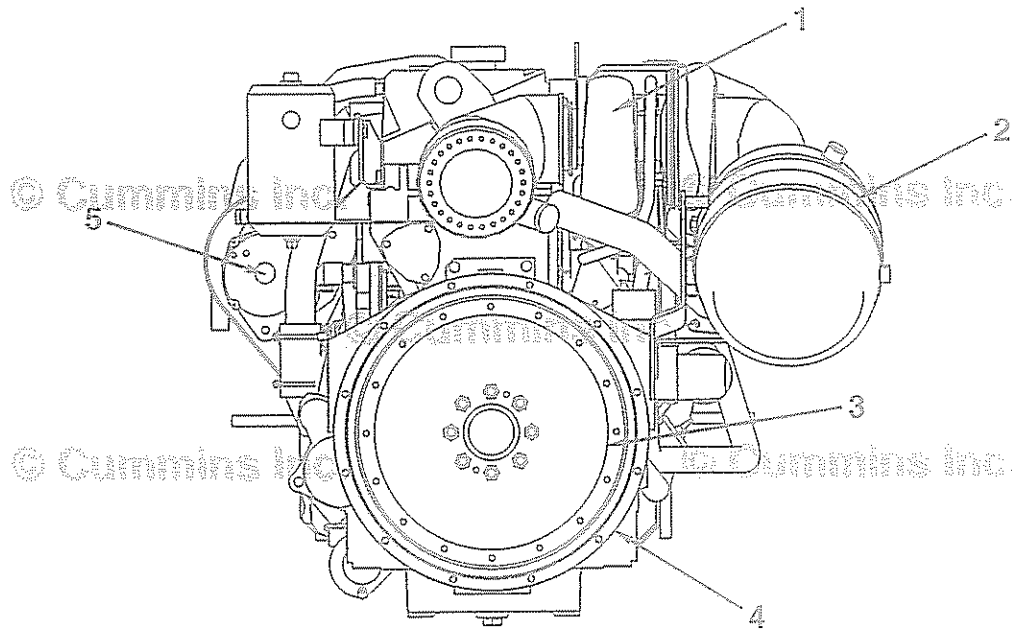


QSM 11 Port View

15d00831

1. Ambient air pressure sensor
2. Oil pressure/temperature sensor (behind fuel pump)
3. Engine position sensor (behind fuel pump)

Marine Applications



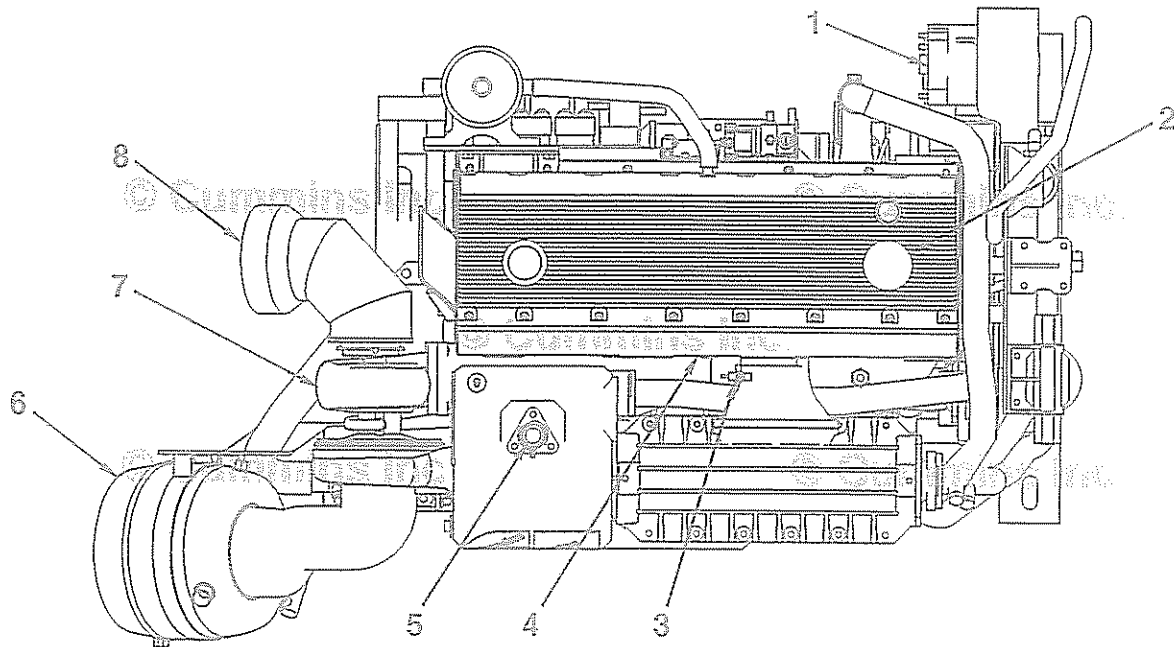
Marine - Rear View

- 1. Turbocharger
- 2. Air cleaner
- 3. Flywheel

- 4. Flywheel housing
- 5. Alternator.

17200017

Marine Applications

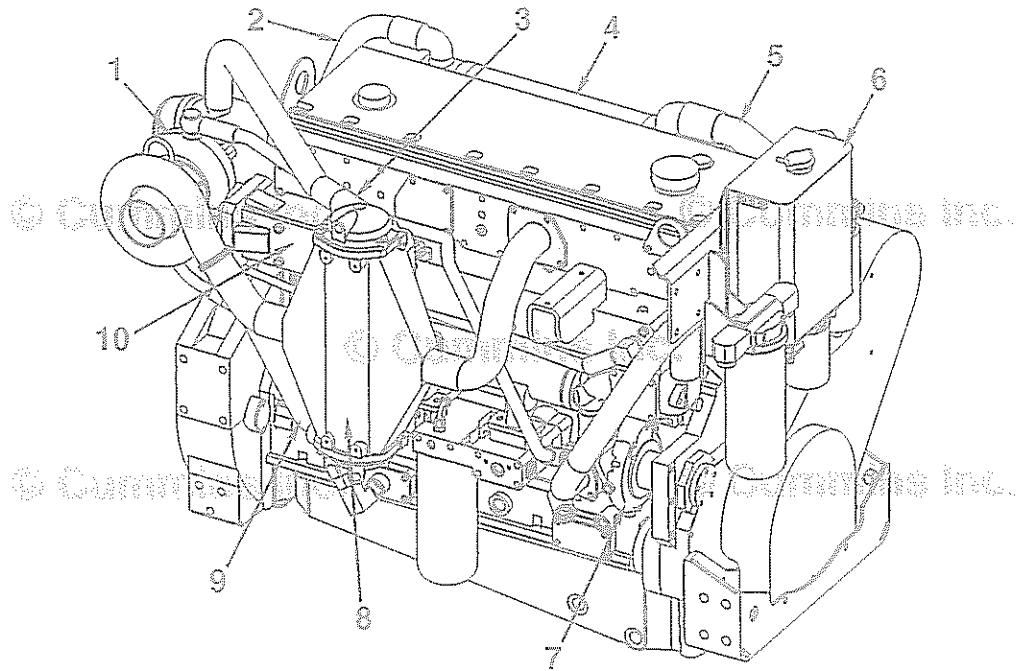


17200016

Marine - Top View

- | | |
|---------------------------------------|-------------------|
| 1. Alternator | 5. Coolant fill |
| 2. Oil fill (alternate) | 6. Air cleaner |
| 3. Intake manifold temperature sensor | 7. Turbocharger |
| 4. Intake manifold pressure sensor | 8. Exhaust elbow. |

Marine Applications

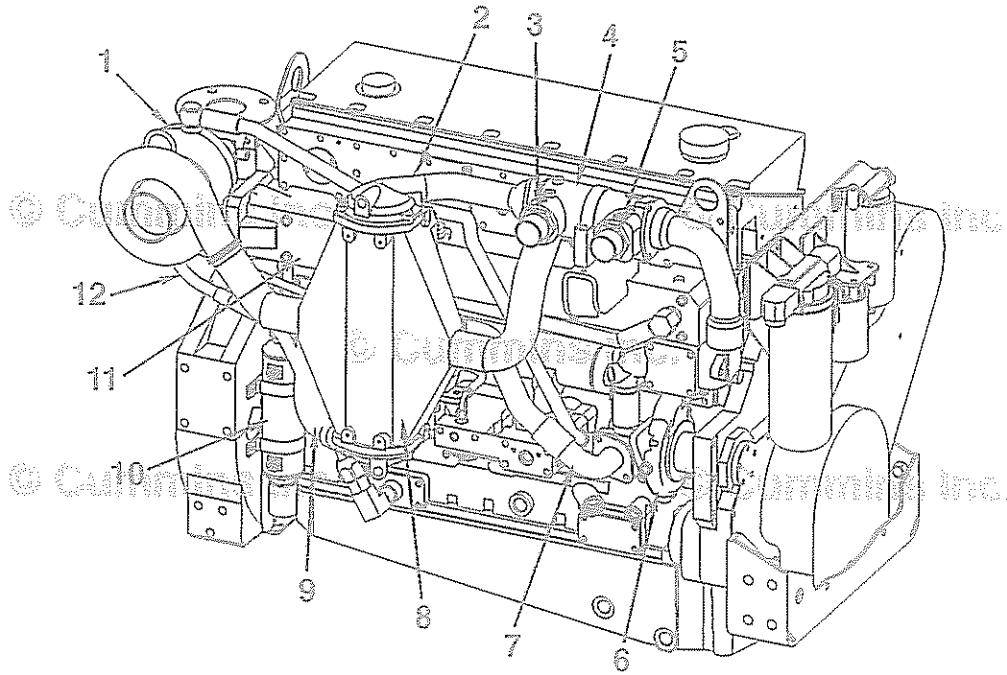


Heat Exchanger

1. Turbocharger with water cooled turbine housing
2. Coolant flow from heat exchanger
3. Aftercooler sea water outlet
4. Heat exchanger
5. Coolant flow to heat exchanger
6. Expansion tank
7. Water pump
8. Aftercooler
9. Aftercooler sea water inlet
10. Water cooled exhaust manifold

08206244

Marine Applications



08200245

1. Turbocharger with water cooled turbine housing
2. Coolant inlet (aftercooler)
3. Coolant in from Keel Cooler
4. LTA thermostat housing
5. Coolant out to Keel Cooler
6. Water pump
7. Coolant to water pump
8. Aftercooler
9. Coolant outlet (aftercooler) to water pump
10. Gear oil cooler (optional)
11. Water cooled exhaust manifold
12. Coolant out from turbocharger

General Engine

Specifications

Industrial and Generator Drive

Type.....Four-cycle, In-line six-cylinder, four-valves per cylinder
Horsepower.....Refer to engine dataplate
Engine Speed at Maximum Output.....Refer to engine dataplate
Bore and Stroke.....125 mm [4.921 in] x 147 mm [5.787 in]
Displacement.....10.8 liters [659 C.I.D.]
Firing Order.....1-5-3-6-2-4
Engine Weight (with standard accessories):.....940 kg [2072 lb]

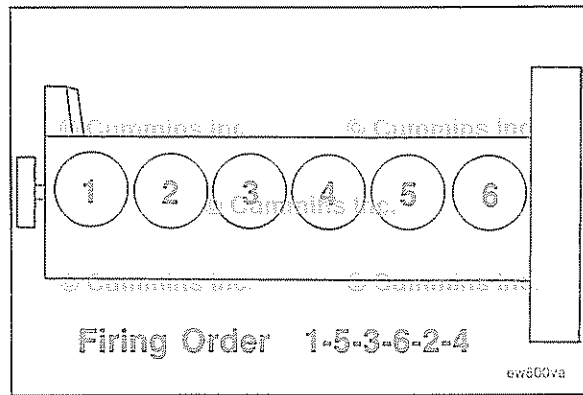
Industrial Applications

Generator Drive Applications

Crankshaft Rotation (viewed from the front of the engine)..... Clockwise

Marine Applications

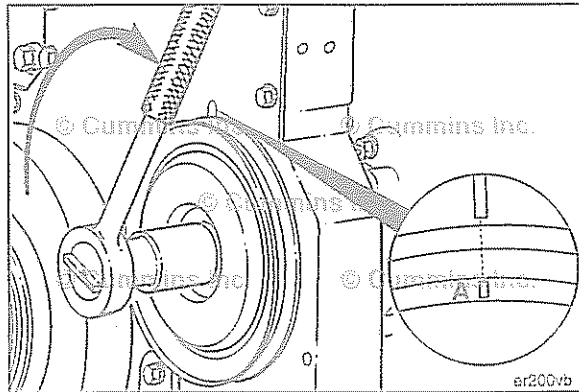
Type.....Four-cycle, In-line six-cylinder, four-valves per cylinder
Horsepower.....Refer to engine dataplate
Engine Speed at Maximum Output.....Refer to engine dataplate
Bore and Stroke.....125 mm [4.921 in] x 147 mm [5.787 in]
Displacement.....10.8 liters [659 C.I.D.]
Firing Order.....1-5-3-6-2-4
Engine Weight (with standard accessories):.....1124 kg [2478 lb]
Dry Weight.....1124 kg [2478 lb]
Crankshaft Rotation (viewed from the front of the engine)..... Clockwise



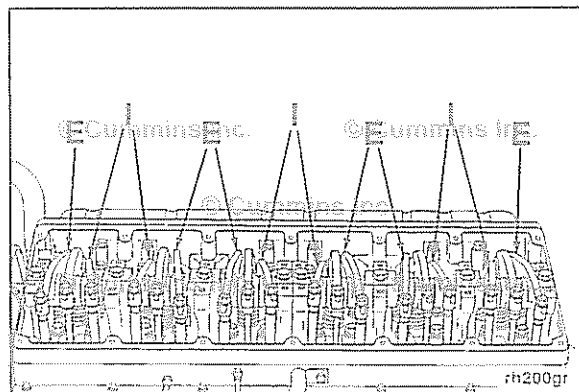
The crankshaft rotation is clockwise when viewed from the front of the engine.

The cylinders are numbered from the front end of the engine.

The firing order is 1-5-3-6-2-4.



Rotate the accessory drive clockwise until the A valve set mark on the accessory drive pulley is aligned with the pointer on the gear cover.



Each cylinder has three rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The center rocker lever is the injector lever
- The short rocker lever (I) is the intake lever.

Exhaust System

Specifications

Marine Applications

Exhaust Pipe Size (normally acceptable inside diameter).....102 mm [4 in]

Maximum Exhaust Pipe Restriction

H₂O.....1016 mm H₂O [40 in H₂O]

Hg.....75 mm Hg [3 in Hg]

Industrial and Generator Drive

Exhaust Pipe Size (normally acceptable inside diameter).....102 mm [4 in]

Maximum Exhaust Pipe Restriction

H₂O.....1016 mm H₂O [40 in H₂O]

Hg.....75 mm Hg [3 in Hg]

Compressed Air System

Specifications

Cummins 18.7 CFM Model

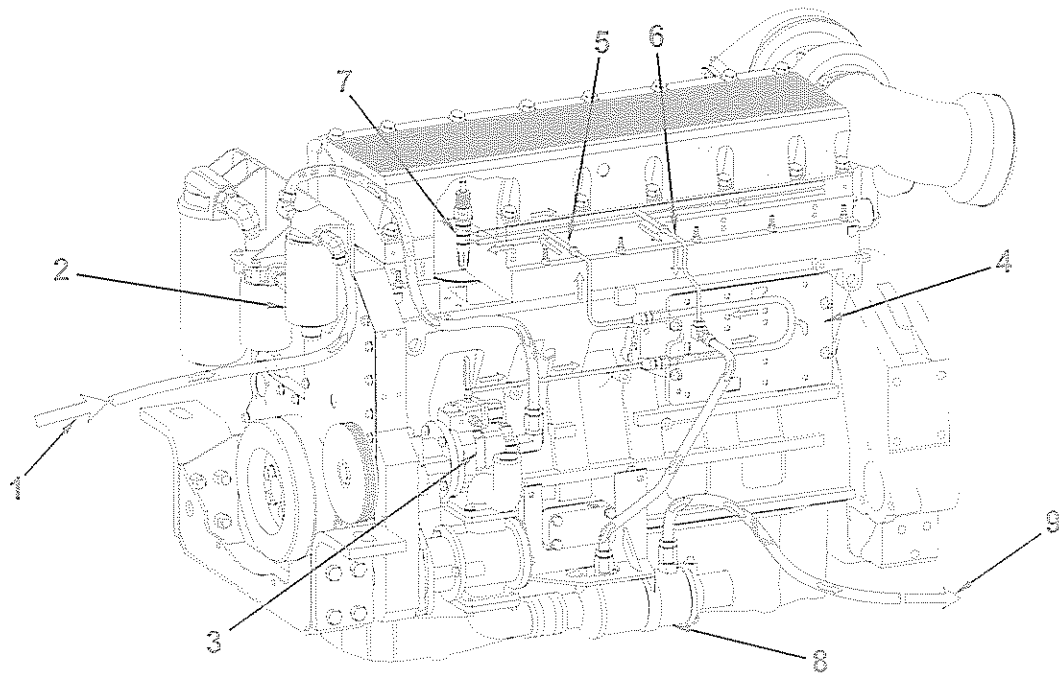
Cylinders.....	1
Compressor Swept Volume at 1250.....	6.6 l/sec. [140 cfm]
Piston Displacement.....	318 cc [19.405 C.I.D.]
Bore.....	85 mm [3.346 in]
Stroke.....	56 mm [2.205 in]
Speed.....	Engine speed
Cooling.....	Engine coolant
Lubrication.....	Engine lubricating oil
Plumbing Line Sizes:	
Coolant Inlet and Outlet.....	3/4 x 16 STOR
Air Inlet.....	1-in nominal hose
Air Outlet.....	M27 x 2 STOR
Unloader Port.....	M10 x 1 STOR
Governor Mounting Direct.....	M8 x .25
Height, Overall (approximate).....	217.4 mm [8.559 in]
Width, Overall (approximate).....	142 mm [5.591 in]
Length, Overall (approximate).....	216 mm [8.504 in]
Weight (approximate).....	15 kg [33.0 lb]

Cummins 37.4 CFM Model

Cylinders.....	1
Compressor Swept Volume at 1250.....	15.5 l/sec. [35.5 cfm]
Piston Displacement.....	636 cc [39 C.I.D.]
Bore.....	85 mm [3.35 in]
Stroke.....	56 mm [2.2 in]
Speed.....	Engine speed
Cooling.....	Engine coolant
Lubrication.....	Engine lubricating oil
Plumbing Line Sizes:	
Coolant Inlet and Outlet.....	3/4 x 16 STOR
Air Inlet.....	1-in nominal hose
Air Outlet.....	M27 x 2 STOR
Unloader Port.....	M10 x 1 STOR
Governor Mounting Direct.....	M8 x .25
Height, Overall (approximate).....	306 mm [12 in]
Width, Overall (approximate).....	142 mm [5.6 in]
Length, Overall (approximate).....	286.7 mm [11.29 in]
Weight (approximate).....	17.24 kg [38 lb]

Flow Diagram, Fuel System

Fuel System Marine Applications

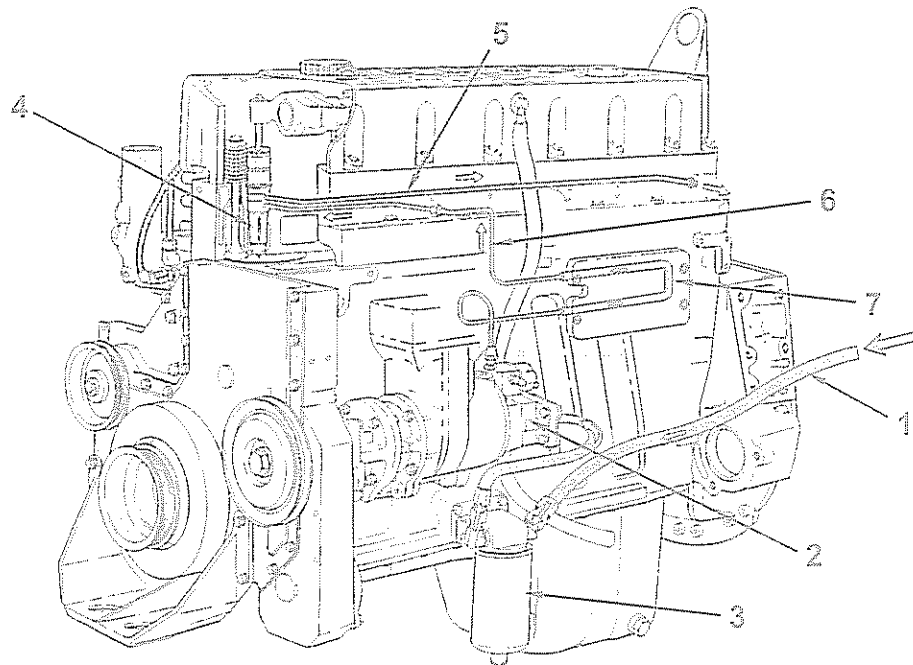


1. Fuel supply from tank
2. Fuel filter
3. Fuel pump
4. ECM cooling plate
5. Fuel to injector

6. Fuel drain to from injector
7. Injector
8. Fuel cooler
9. Fuel return to supply tank

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Industrial and Generator Drive

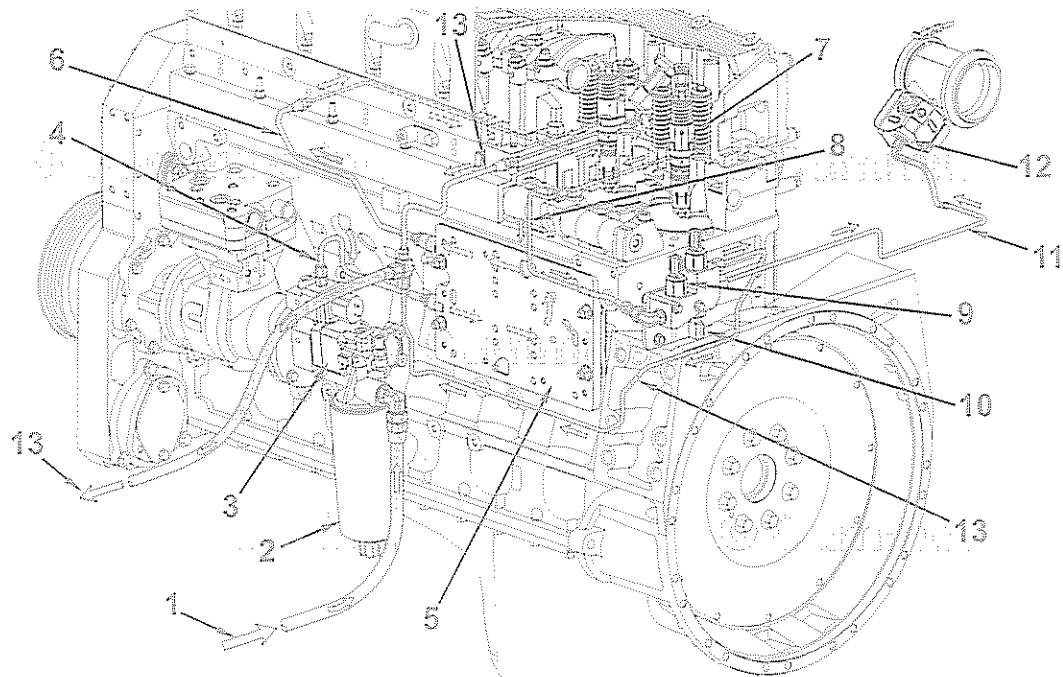


- 1. Fuel supply from tank
- 2. Fuel pump
- 3. Fuel filter
- 4. Injector

- 5. Fuel drain from injector
- 6. Fuel to injector
- 7. ECM cooling plate

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Automotive with CM876

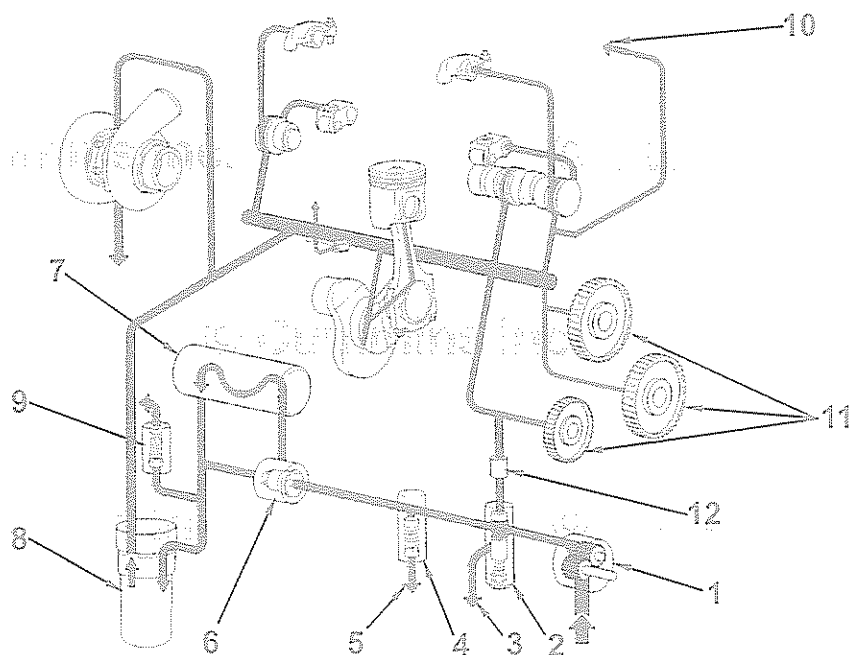


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- | | |
|------------------------------|--|
| 1. Fuel supply from tank | 8. Fuel supply to aftertreatment injector system |
| 2. Fuel filter | 9. Fuel shutoff valve |
| 3. Fuel Pump | 10. Fuel pressure sensor |
| 4. Fuel supply to ECM cooler | 11. Fuel supply to aftertreatment fuel injector |
| 5. ECM cooling plate | 12. Aftertreatment fuel injector |
| 6. Fuel supply to injectors | 13. Fuel drain to tank |
| 7. Injector | |

Flow Diagram, Lubricating Oil System

Flow Diagram

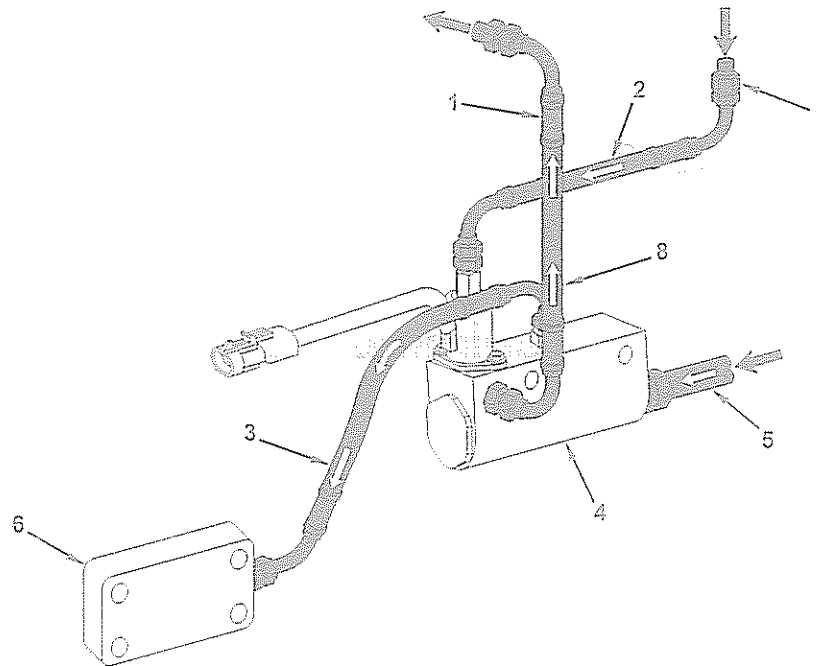


- | | |
|-------------------------------|---------------------------------------|
| 1. Oil pump | 7. Oil cooler |
| 2. Pressure regulator valve | 8. Combination oil filter |
| 3. Oil return to pan | 9. Filter bypass valve |
| 4. High pressure relief valve | 10. Accessory drive or air compressor |
| 5. Oil return to pan | 11. Idler gears |
| 6. Oil thermostat | 12. Viscosity sensor. |

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Flow Diagram, Lubricating Oil System

Flow Diagram



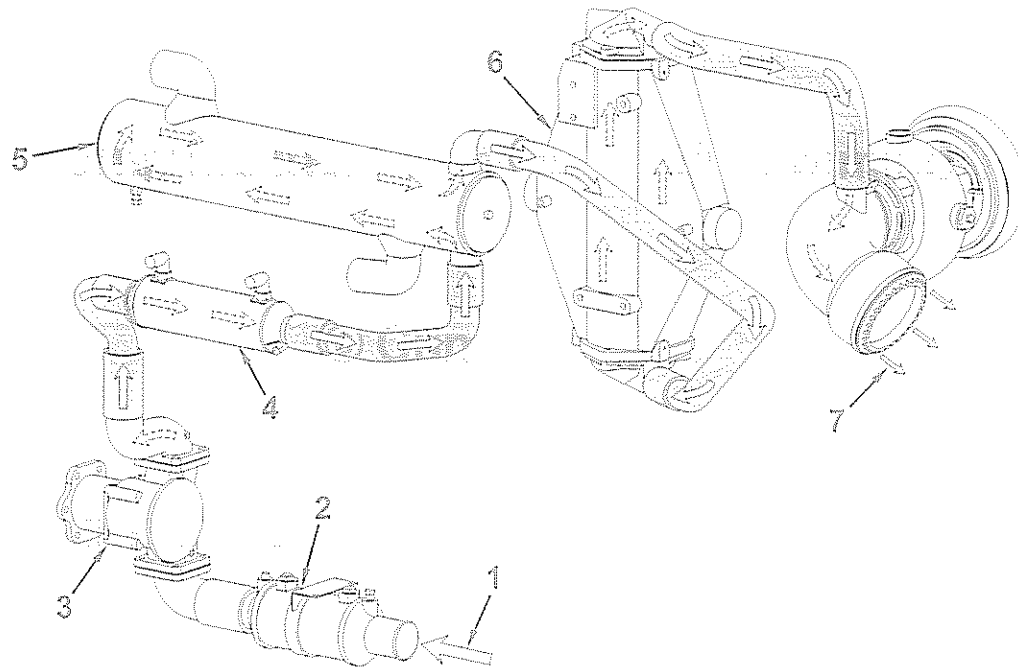
07200086

Centinel™ (optional)

- | | |
|--|--|
| 1. Oil burn line (used oil to fuel return) | 5. OEM fresh oil supply (from makeup tank) |
| 2. Oil rifle hose (used oil inlet) | 6. Handhole cover |
| 3. Oil replenishment hose (clean oil to pan) | 7. Rifle check valve |
| 4. Control valve | 8. Outlet check valve. |

Flow Diagram, Cooling System

Flow Diagram Marine Applications



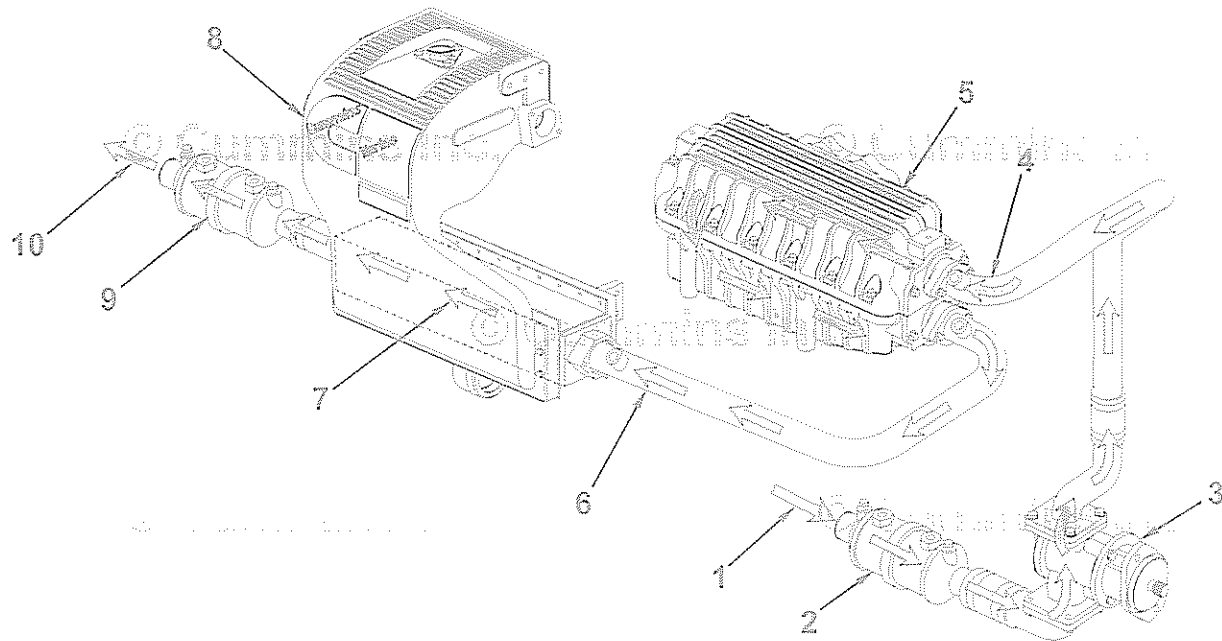
Sea Water Flow (Heat Exchanger)

- | | |
|--------------------|----------------------|
| 1. Sea water inlet | 5. Heat exchanger |
| 2. Fuel cooler | 6. Aftercooler |
| 3. Sea water pump | 7. Sea water outlet. |
| 4. Gear oil cooler | |

09200243

Flow Diagram, Cooling System

Flow Diagram Marine Applications



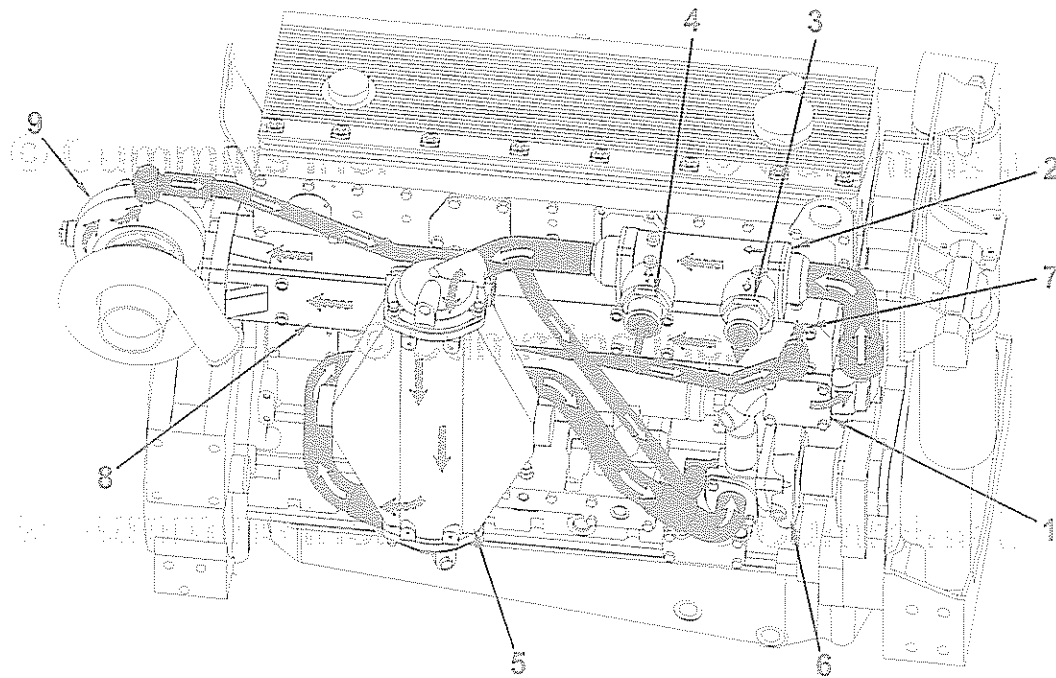
08200150

Sea Water Flow (Heat Exchanger)

- | | |
|----------------------------------|----------------------------------|
| 1. Sea water inlet | 6. Aftercooler to heat exchanger |
| 2. Fuel cooler | 7. Heat exchanger |
| 3. Sea water pump | 8. Expansion tank |
| 4. Sea water pump to aftercooler | 9. Gear oil cooler |
| 5. Aftercooler | 10. Sea water outlet. |

Flow Diagram, Cooling System

Flow Diagram Marine Applications



Coolant Flow, Keel Cooled (one of three)

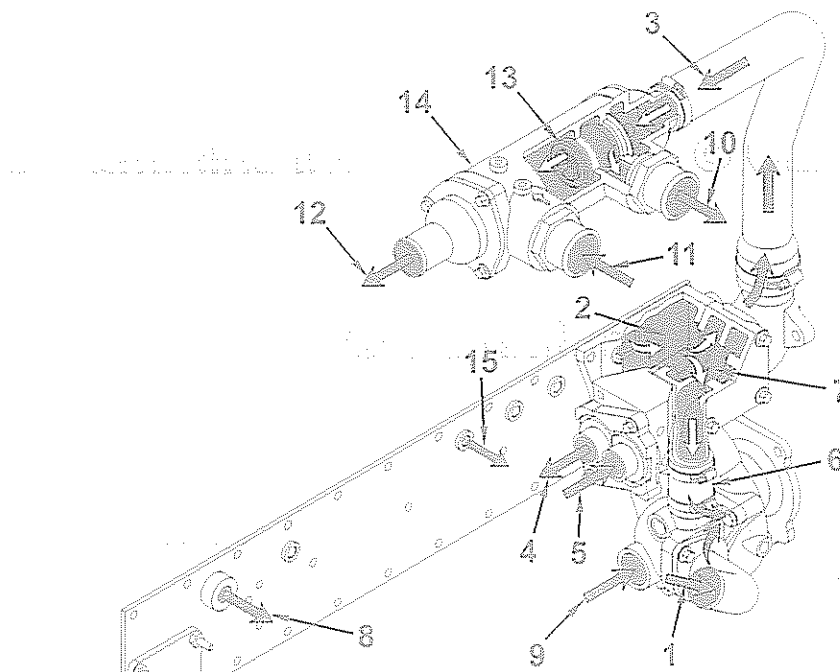
- | | |
|--|-------------------------------------|
| 1. Engine thermostat cavity (thermostat removed) | 6. Water pump |
| 2. Keel cooled thermostat housing | 7. Coolant flow to exhaust manifold |
| 3. Coolant out to keel cooler | 8. Water cooled exhaust manifold |
| 4. Coolant in from keel cooler | 9. Water cooled turbine housing. |
| 5. Aftercooler | |

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Flow Diagram, Cooling System

Flow Diagram

Marine Applications



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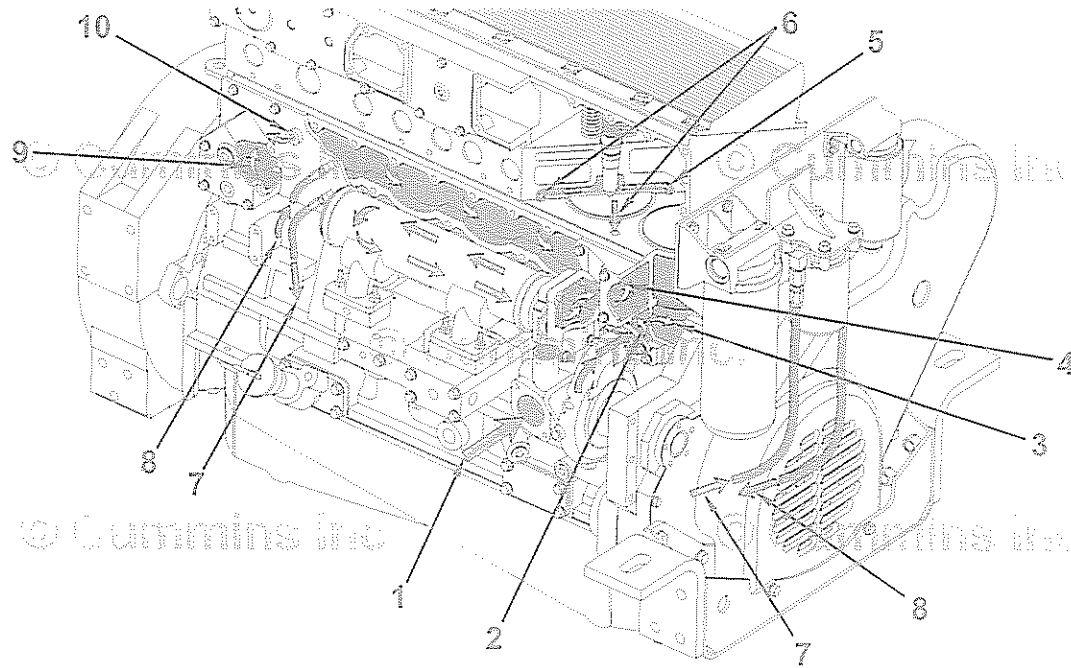
Coolant Flow, Keel Cooled (two of three)

- | | |
|--|--|
| 1. Water pump coolant inlet | 9. Turbocharger coolant return to water pump inlet |
| 2. Upper manifold cavity | 10. To keel cooler |
| 3. Coolant outlet | 11. From keel cooler |
| 4. Lower manifold cavity to cooler | 12. To aftercooler |
| 5. Cooler to upper manifold cavity (before thermostat) | 13. LTA thermostat |
| 6. Coolant bypass hose - plugged | 14. LTA thermostat housing |
| 7. Engine thermostat cavity (thermostat removed) | 15. To turbocharger. |
| 8. To exhaust manifold | |

Flow Diagram, Cooling System

Flow Diagram

Marine Applications



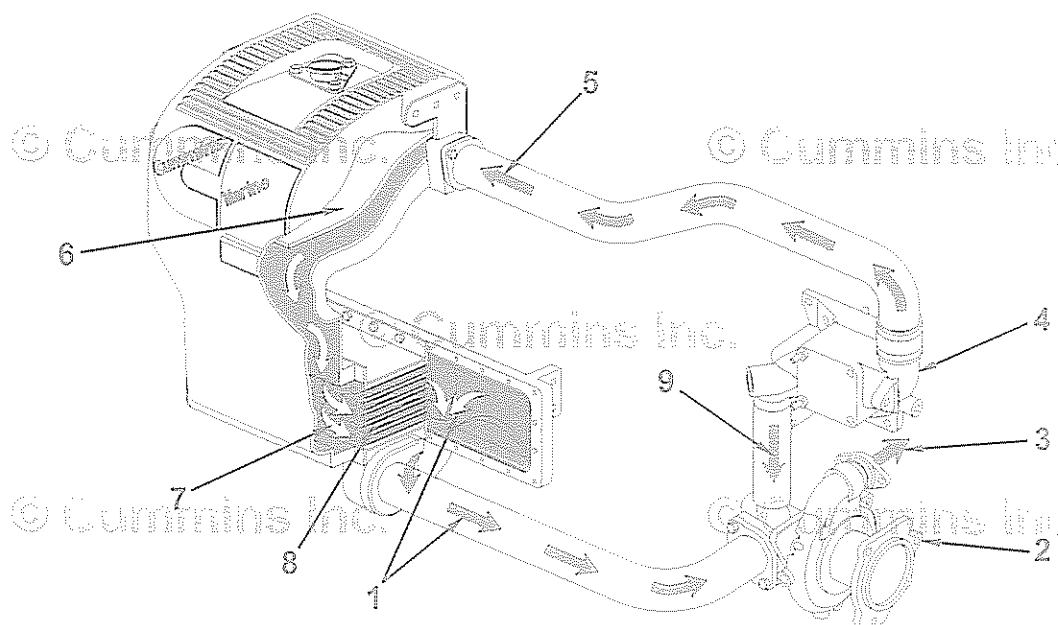
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Coolant Flow (three of three)

- | | |
|---|---|
| 1. Flow to water pump | 6. Flow from cylinder head to upper manifold cavity |
| 2. Flow to oil cooler and lower manifold cavity | 7. Flow to coolant filter |
| 3. Flow from lower manifold cavity to cylinder liners | 8. Flow from coolant filter |
| 4. Flow from oil cooler to thermostat housing | 9. Coolant heater - optional |
| 5. Flow to cylinder head | 10. Upper manifold to exhaust manifold. |

Flow Diagram, Cooling System

Flow Diagram Marine Applications



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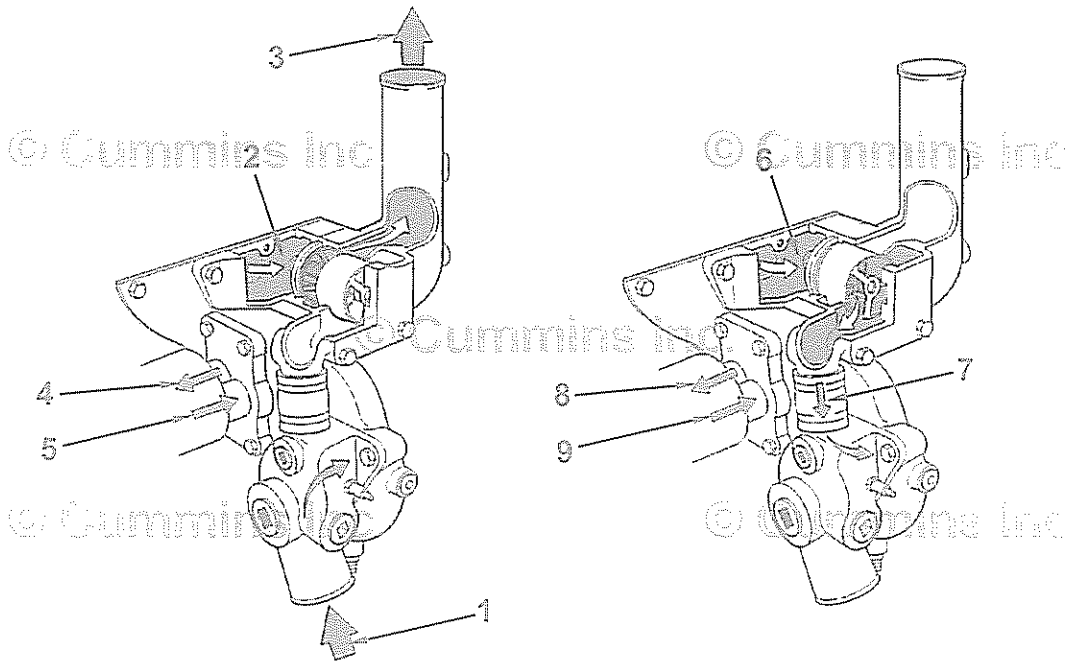
Coolant Flow, Heat Exchanger (one of three)

- | | |
|--|--|
| 1. Coolant flow to water pump | 6. Expansion tank |
| 2. Water pump | 7. Heat exchanger sea water core |
| 3. Coolant flow to lower manifold cavity | 8. Coolant flow through sea water core |
| 4. Thermostat housing | 9. Coolant bypass. |
| 5. Thermostat housing to heat exchanger | |

Flow Diagram, Cooling System

Flow Diagram

Marine Applications



Coolant Flow, Heat Exchanger (two of three)

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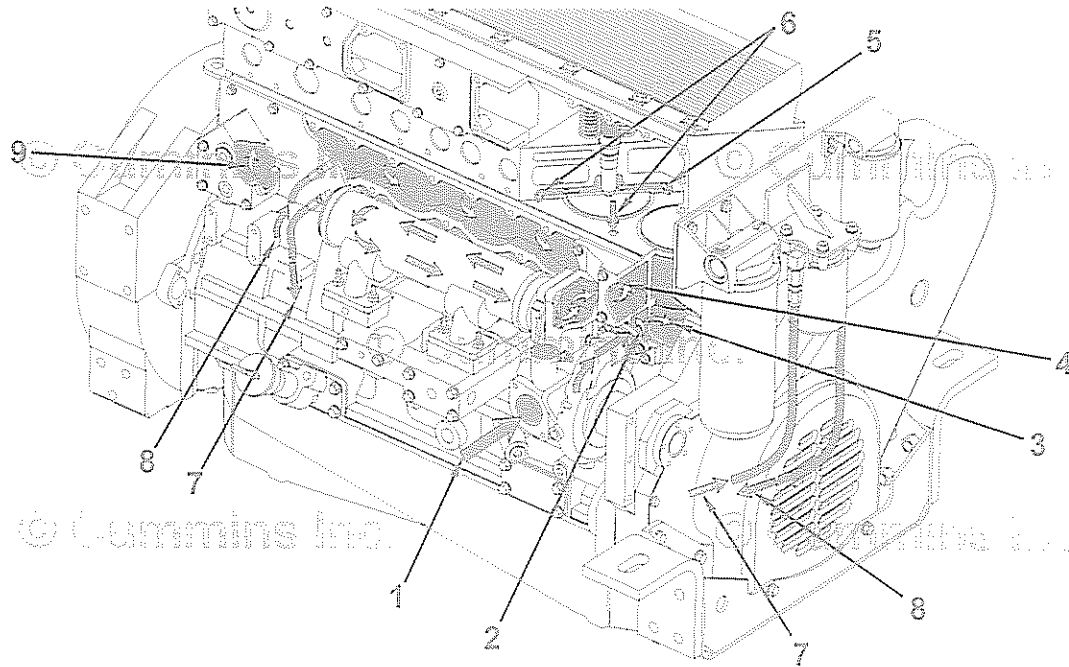
Thermostat Open (1-5)

Thermostat Closed (6-9)

- | | |
|---|---|
| 1. Water pump coolant inlet | 6. Upper manifold cavity (coolant to thermostat) |
| 2. Upper manifold cavity | 7. Coolant bypass (return to water pump) |
| 3. Coolant outlet | 8. Lower manifold cavity to cooler |
| 4. Lower manifold cavity to cooler | 9. Cooler to upper manifold cavity (before thermostat). |
| 5. Cooler to upper manifold cavity (before thermostat). | |

Flow Diagram, Cooling System

Flow Diagram Marine Applications

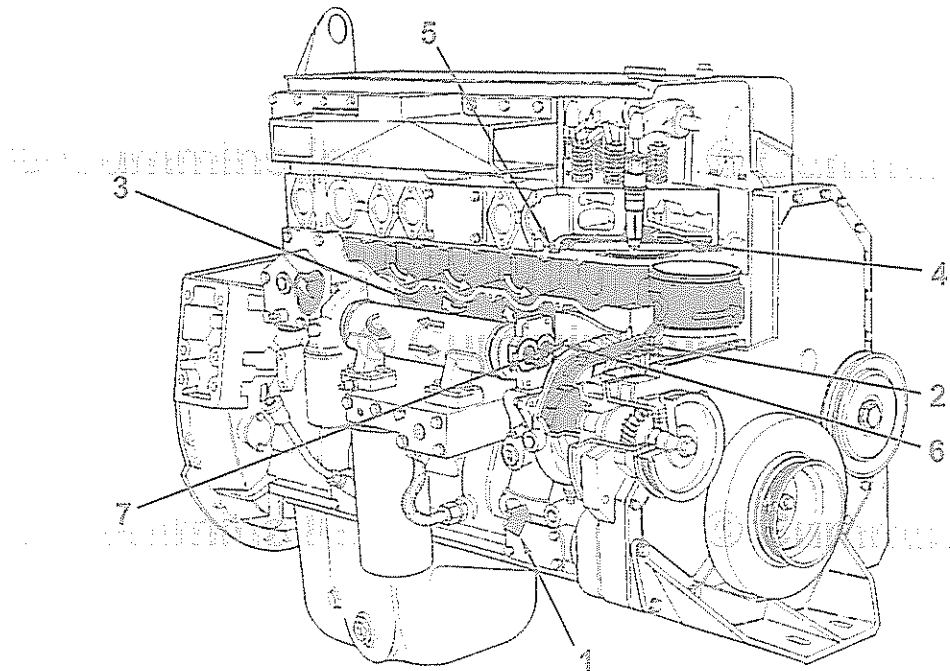


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Coolant Flow, Heat Exchanger (three of three)

- | | |
|---|---|
| 1. Flow from heat exchanger to water pump | 6. Flow from cylinder head to upper manifold cavity |
| 2. Flow to oil cooler and lower manifold cavity | 7. Flow to coolant filter |
| 3. Flow from lower manifold cavity to cylinder liners | 8. Flow from coolant filter |
| 4. Flow from oil cooler to thermostat housing | 9. Coolant heater - optional. |
| 5. Flow to cylinder head | |

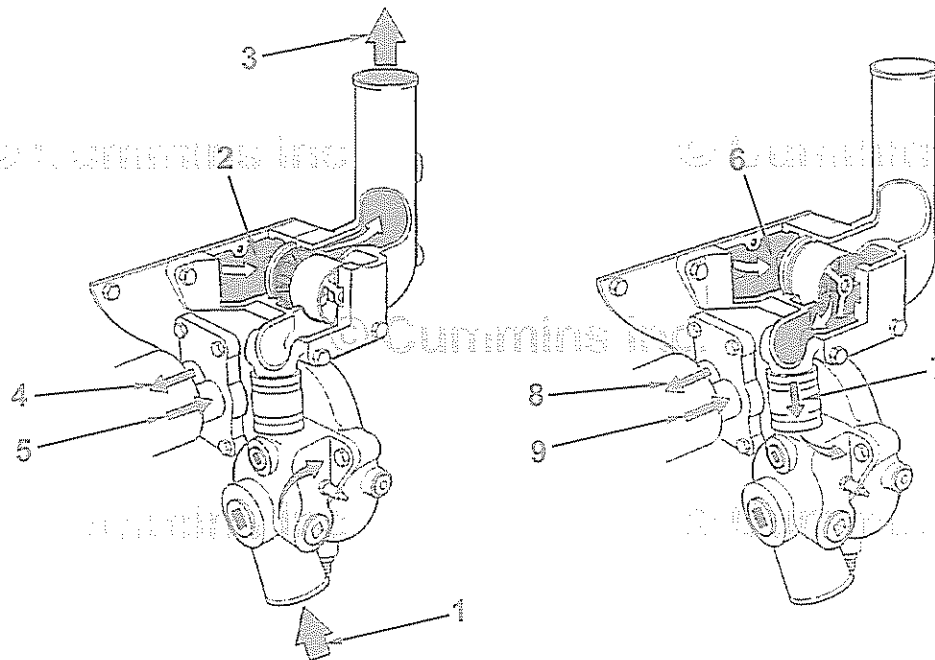
Industrial and Generator Drive



- | | |
|---|---|
| 1. Water pump coolant inlet | 5. Coolant to upper manifold cavity |
| 2. Coolant to lower manifold cavity | 6. Lower manifold coolant to oil cooler |
| 3. Coolant to cylinder liner block cavity | 7. Oil cooler to upper manifold cavity. |
| 4. Coolant to cylinder head | |

08200086

Industrial and Generator Drive



03200087

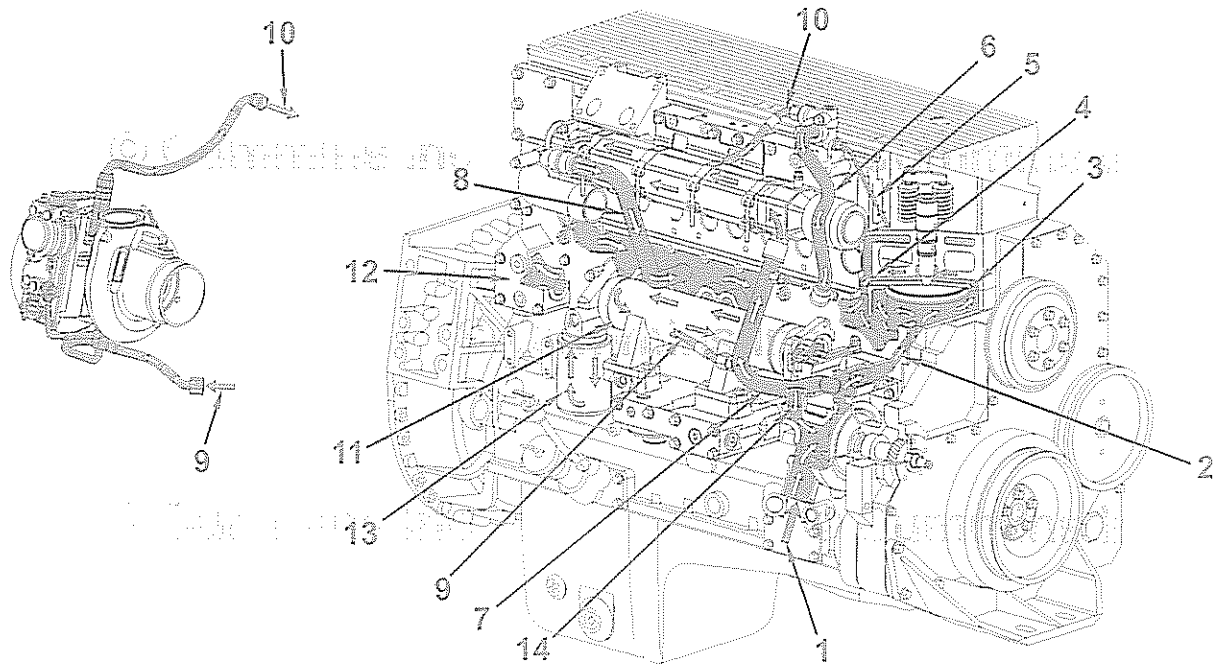
Open (1-5)

Closed (6-9)

1. Water pump coolant inlet
2. Upper manifold cavity (coolant to thermostat)
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat).

6. Upper manifold cavity (coolant to thermostat)
7. Coolant bypass (return to water pump)
8. Lower manifold cavity to cooler
9. Cooler to upper manifold cavity (before thermostat).

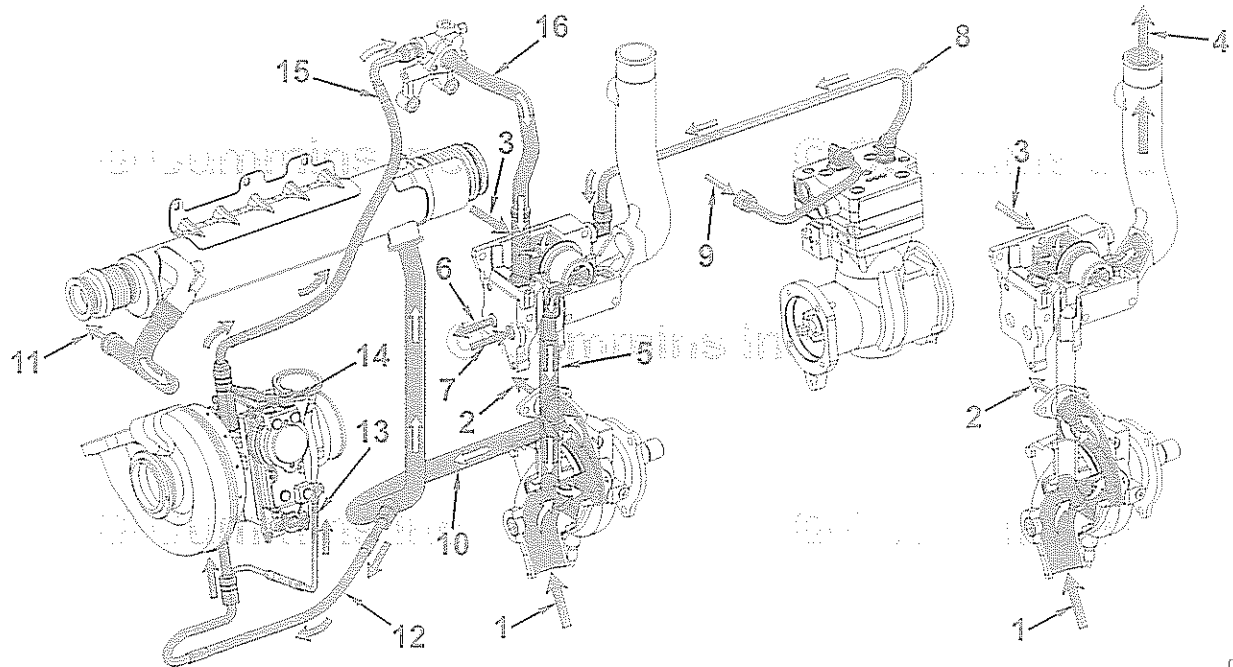
Automotive with CM876



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- | | |
|---|---|
| 1. Water pump coolant inlet | 8. EGR cooler coolant return to low pressure cavity |
| 2. Coolant supply to block | 9. Coolant supply to turbocharger |
| 3. Coolant supply to cylinder head | 10. Turbocharger coolant return to water manifold |
| 4. Coolant return to low pressure cavity | 11. Oil cooler |
| 5. Cylinder head coolant vent to water manifold | 12. Heater housing |
| 6. Coolant return to thermostat housing | 13. Coolant filter |
| 7. Coolant supply to EGR cooler | 14. Coolant return from thermostat housing. |

Automotive with CM876

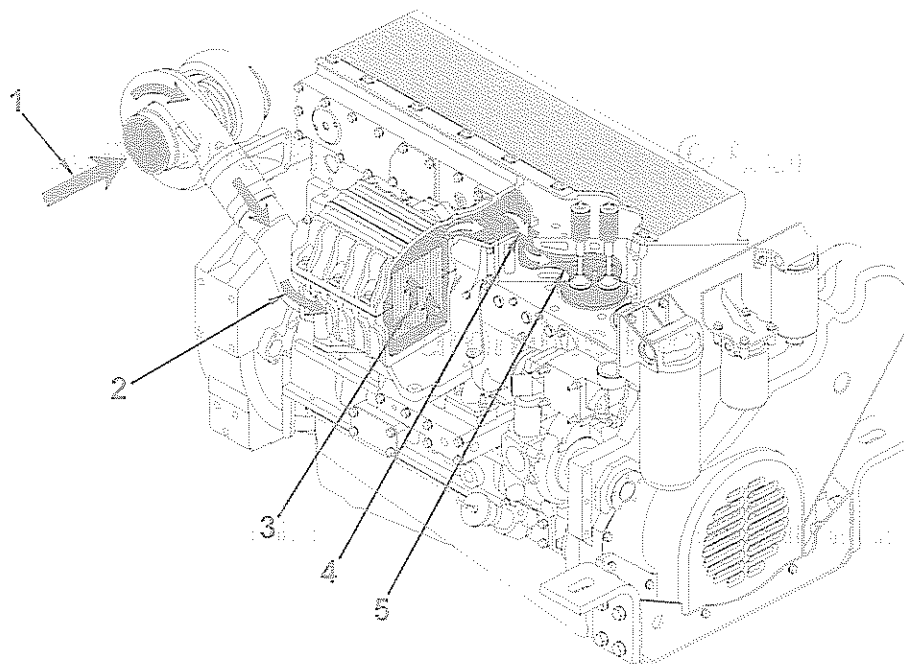


08260248

- | | |
|--|---|
| 1. Water pump coolant inlet | 9. Coolant to air compressor |
| 2. Coolant to block | 10. Coolant supply to EGR cooler |
| 3. Upper manifold cavity (coolant to thermostat) | 11. Coolant from EGR cooler to block |
| 4. Coolant outlet (coolant to radiator) | 12. Coolant supply to turbocharger |
| 5. Coolant bypass (return to water pump) | 13. Coolant supply to turbocharger actuator |
| 6. Lower manifold cavity to lubricating oil cooler | 14. Coolant return from turbocharger actuator |
| 7. Lubricating oil cooler to upper manifold cavity (before thermostat) | 15. Coolant from turbocharger to water manifold |
| 8. Coolant from air compressor | 16. Coolant return from water manifold to thermostat housing (before thermostat). |

Flow Diagram, Air Intake System

Flow Diagram Marine Applications

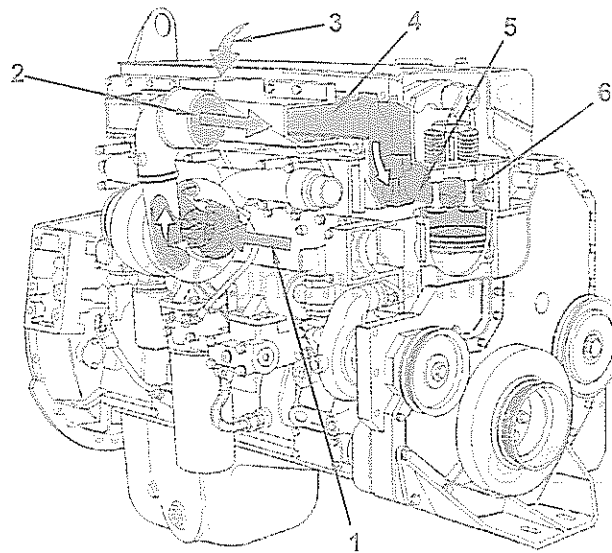


1. Filtered intake air to turbocharger
2. Turbocharger air to aftercooler
3. Intake air through the aftercooler

4. Aftercooled air through intake manifold
5. Intake valve ports.

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Industrial and Generator Drive

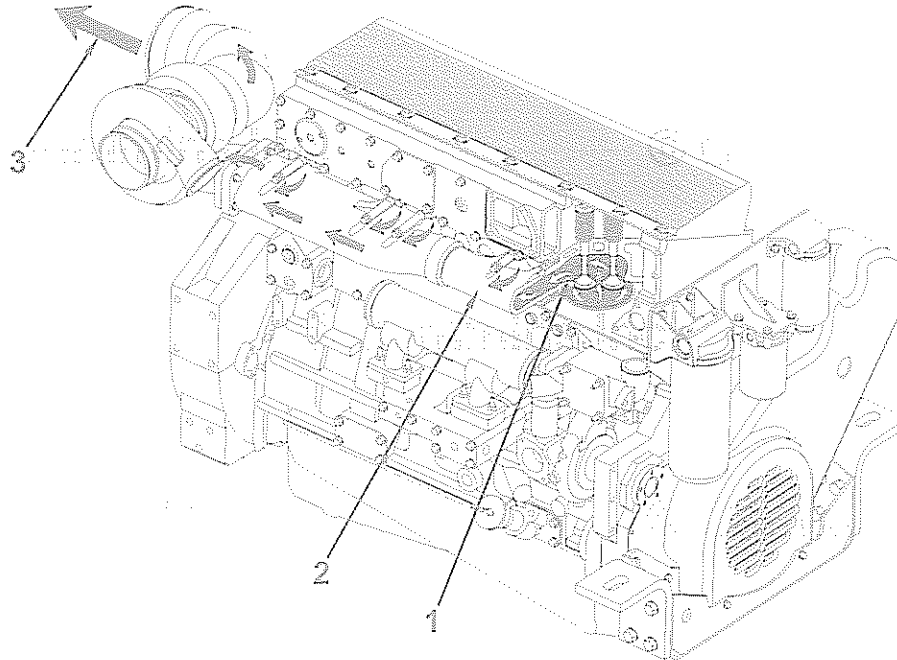


10200035

- | | |
|--|-----------------------|
| 1. Filtered intake air to turbocharger | 4. Intake manifold |
| 2. Turbocharger air to charge air cooler | 5. Intake valve ports |
| 3. Charge air cooler intake air to intake manifold | 6. Intake valves. |

Flow Diagram, Exhaust System

Flow Diagram Marine Applications

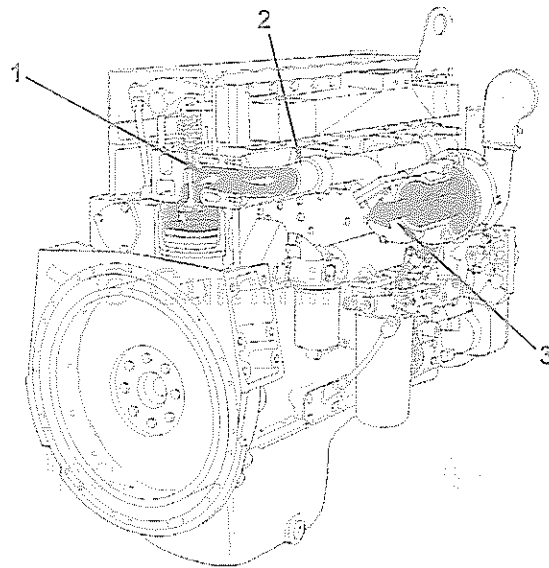


- 1. Exhaust valve ports
- 2. Exhaust manifold

- 3. Turbocharger exhaust outlet.

11200027

Industrial and Generator Drive

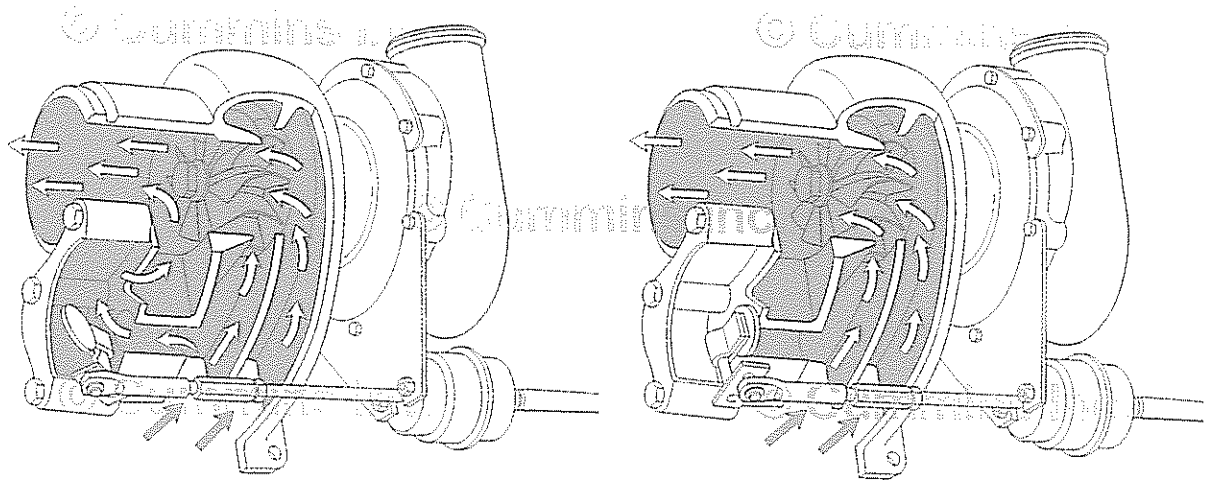


- 1. Exhaust valve ports
- 2. Exhaust manifold

- 3. Turbocharger exhaust outlet.

11200009

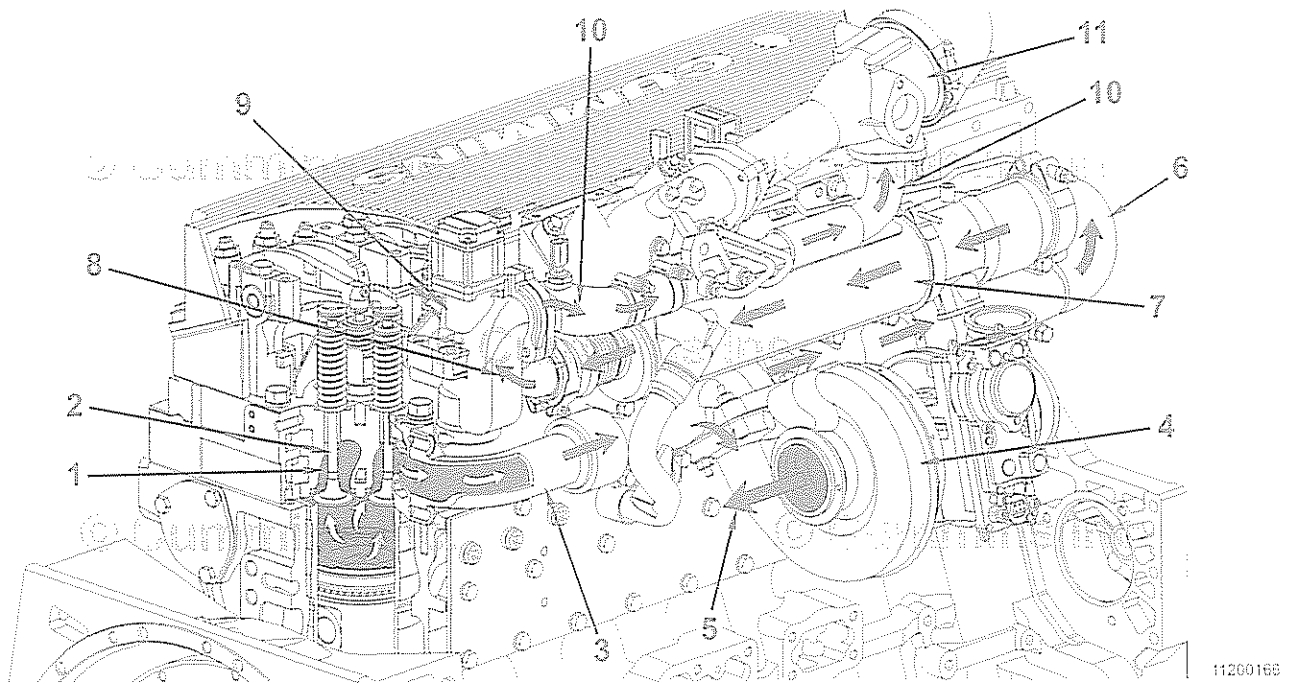
Industrial and Generator Drive



Wastegate Actuation OPEN and CLOSED

12200015

Automotive with CM876

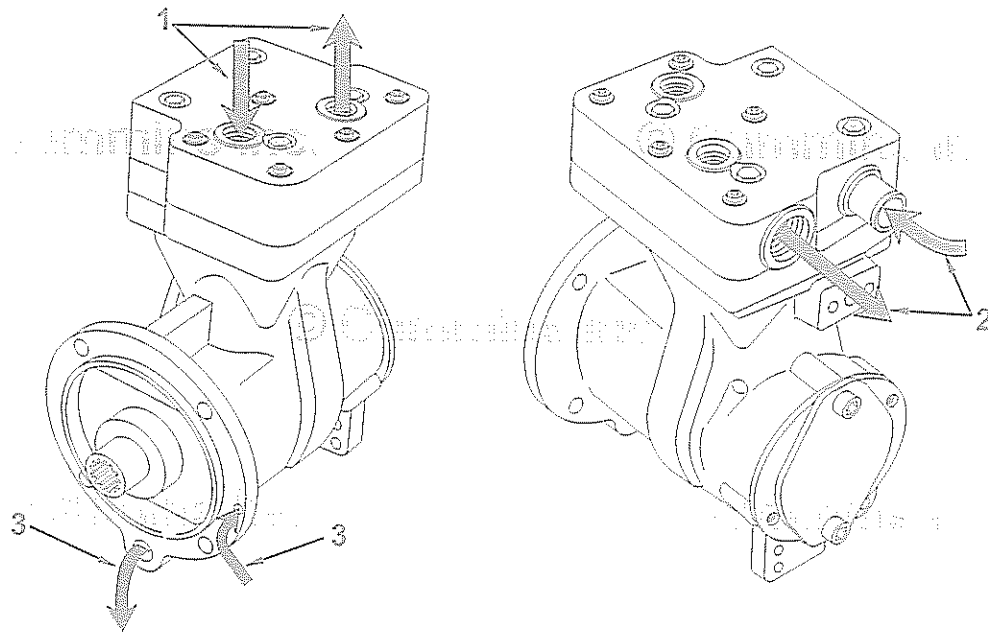


1. Exhaust gas to manifold
2. Exhaust valve
3. Exhaust gas to turbocharger
4. Variable geometry turbocharger
5. Turbocharger exhaust gas outlet
6. Exhaust gas to EGR cooler

7. EGR cooler
8. Cooled exhaust gas to EGR valve
9. EGR valve
10. Cooled exhaust gas from the EGR valve to the EGR mixer
11. EGR mixer

Flow Diagram, Compressed Air System

Compressed Air System
Cummins

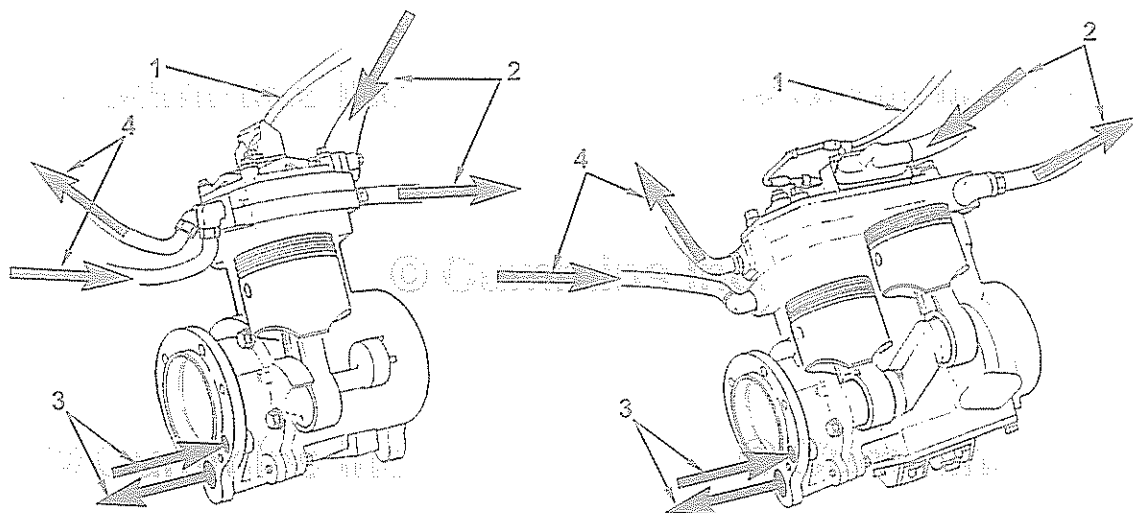


- 1. Coolant
- 2. Air

- 3. Lubricant.

10200072

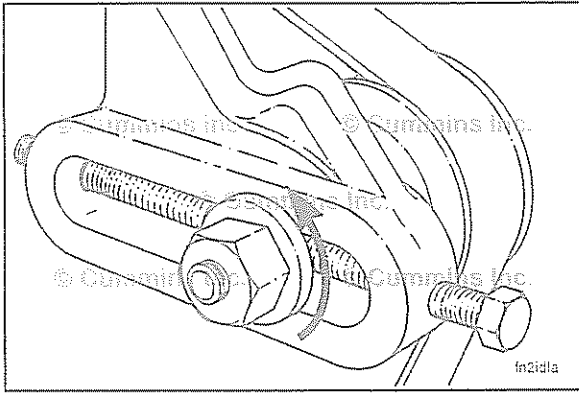
Holset® Models



- 1. Air governor signal
- 2. Air

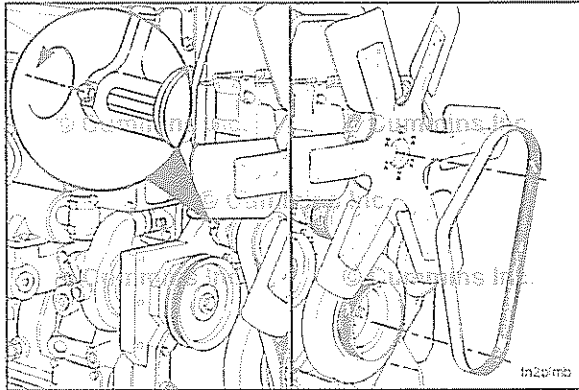
- 3. Lubricant
- 4. Coolant.

12200009



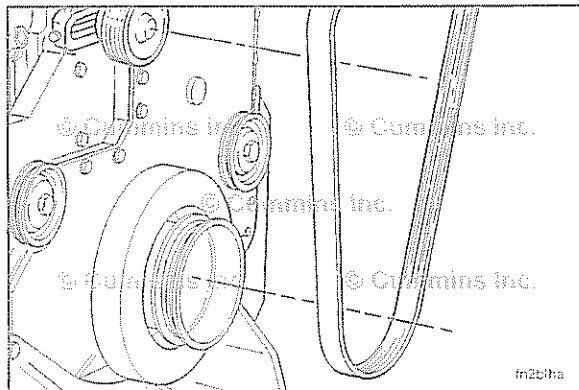
Remove

Loosen the fan idler pulley shaft locknut.



Turn the adjusting screw counterclockwise to release the belt tension.

Move the fan idler pulley and fan pulley centers as close as possible. The belt can then be removed without excessive force.

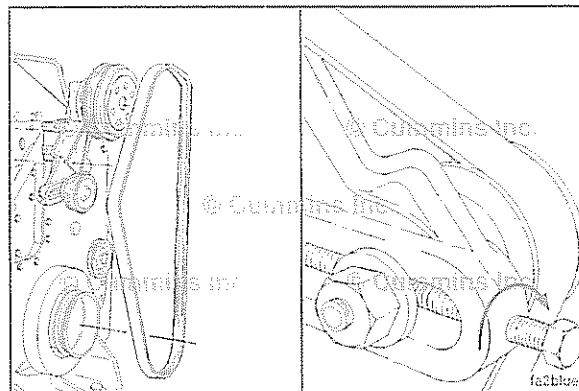


Install

⚠CAUTION⚠

To reduce the possibility of damage to the pulley and new belt, do not roll the belt over the pulley or pry it on with a tool.

Install a new fan belt on the pulleys.



Use belt tension gauge, Part Number ST-1293, or equivalent, to measure the belt tension. Refer to Procedure 018-005 (Drive Belt Tension) in Section V.



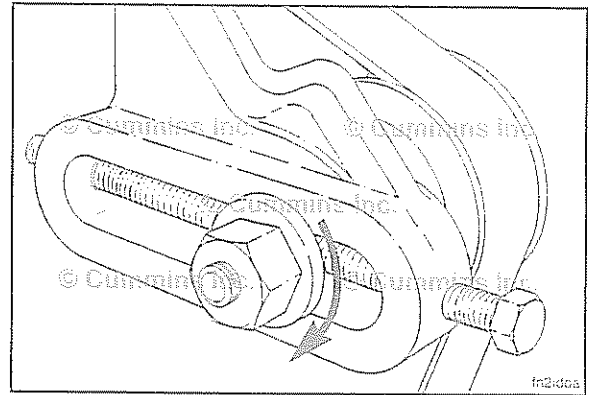
Turn the adjusting screw to adjust the belt tension.

NOTE: A belt is considered used if it has been in operation 10 minutes or longer. If used belt tension is below the minimum value, tighten the belt to the maximum value. Replace the belt if it will not maintain correct tension.

Tighten the idler pulley shaft locknut.

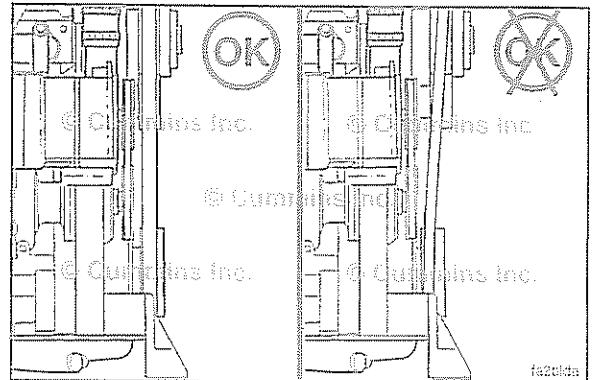
Torque Value: 190 N•m [140 ft-lb]

Check the belt tension again to make sure belt is not too tight.



Inspect the pulley misalignment must not exceed 6 mm for each meter [1/16 inch for each 12 inches] of distance between the pulley centers.

Replace the fan idler pulley and fan hub pulley if they are not in alignment. Refer to Procedure008-030 (Fan Drive Idler Pulley Assembly) in Section A.



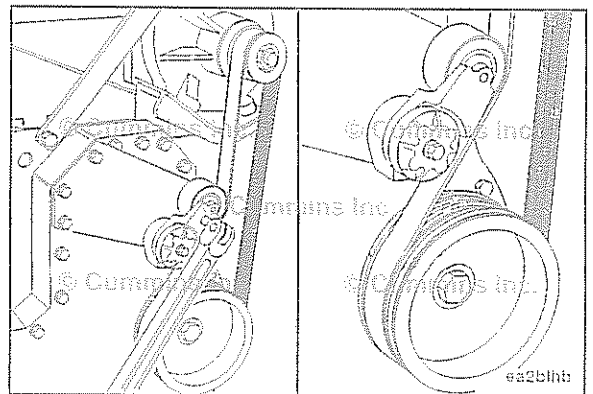
Drive Belt, Alternator

Install

Install a new belt over the pulleys while holding the tensioner back. Be careful not to damage the belt while working it over the flanged pulleys.

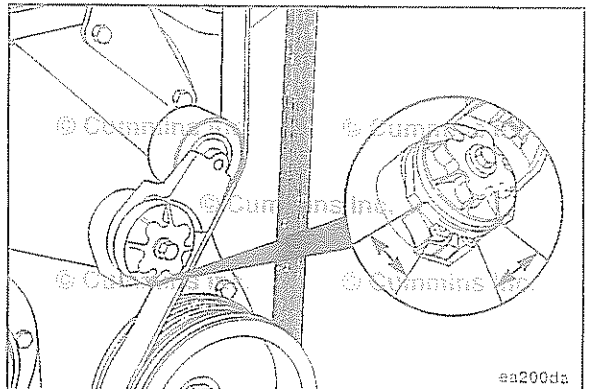
Release the tensioner, and remove the breaker bar.

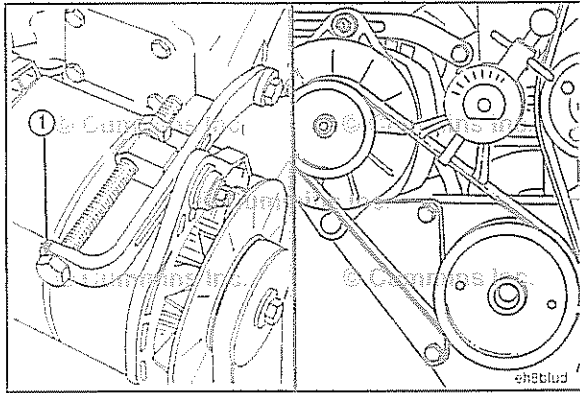
Belt drive systems equipped with an automatic belt tensioner can not be adjusted. A belt tension gauge will not give an accurate measure of the belt tension. The automatic belt tensioner is designed to maintain the proper belt tension over the life of the belt. Only inspection of the tensioner is required.



The belt tensioner is designed to operate within the limit of arm movement provided by the cast stops when the belt length and geometry are correct.

If the tensioner is hitting either of the limits during operation, check the mounting brackets and belt length. Loose brackets, bracket failure, alternator movement, incorrect belt length, or belt failure can cause the tensioner to hit the limits.





Install a new belt on the water pump and alternator pulleys. To prevent damage, do not roll a belt over the pulley or pry on it with a tool.

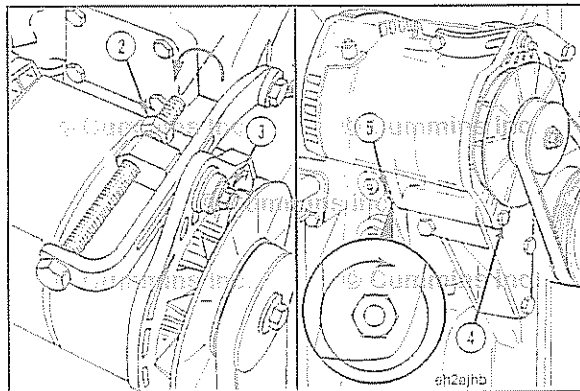


Turn the adjusting screw (1) clockwise to increase the belt tension.



Use belt tension gauge, Part Number ST-1293, or equivalent, to measure the belt tension. Refer to Procedure 018-005 (Drive Belt Tension) in Section V for the correct tension value for the belt is installed.

NOTE: A belt is considered used if it has been in operation for ten minutes or longer.



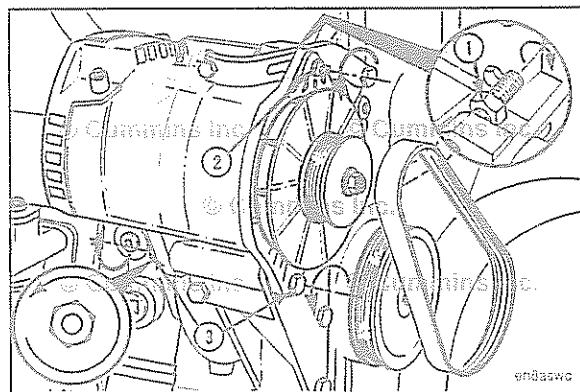
Tighten the adjusting screw locknut (2) against the retainer.

Tighten the adjustment link locking capscrew (3).

Torque Value: 80 N•m [60 ft-lb]

Tighten the pivot capscrew (4) and nut (5).

Torque Value: 47 N•m [35 ft-lb]



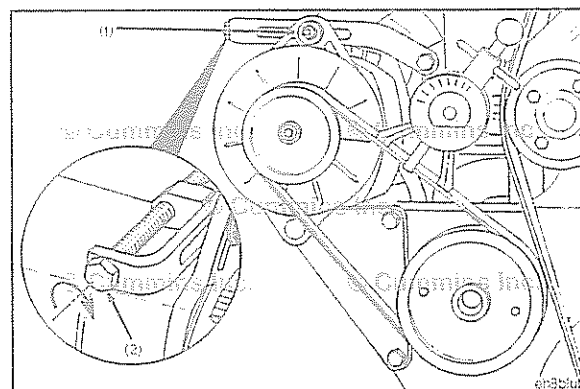
Adjust

Adjustment Link Type

Loosen the adjusting screw locknut (1).

Loosen the adjustment link locking capscrew (2).

Loosen the pivot capscrew and nut (3).



Use belt tension gauge, Part No. ST-1293, to measure belt tension.



Turn the alternator adjusting screw (1) clockwise to tighten the belt. Refer to Procedure 018-005 (Drive Belt Tension) in Section V.

NOTE: A belt is considered used if it has been in operation for ten minutes or longer.

Tighten the adjusting screw locknut (2) against the retainer.

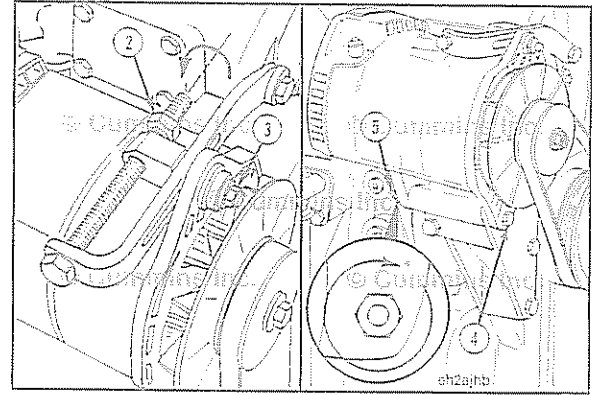


Tighten the adjustment link locking capscrew (3).

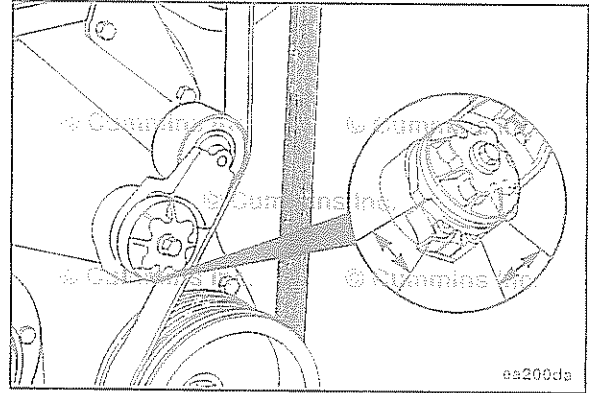
Torque Value: 80 N•m [60 ft-lb]

Tighten the pivot capscrew (4) and nut (5).

Torque Value: 47 N•m [35 ft-lb]



Belt drive systems equipped with an automatic belt tensioner can not be adjusted. A belt tension gauge will not give an accurate measure of the belt tension. The automatic belt tensioner is designed to maintain proper belt tension over the life of the belt. Only inspection of the tensioner is required.



The belt tensioner is designed to operate within the limit of arm movement provided by the cast stops when the belt length and geometry are correct.

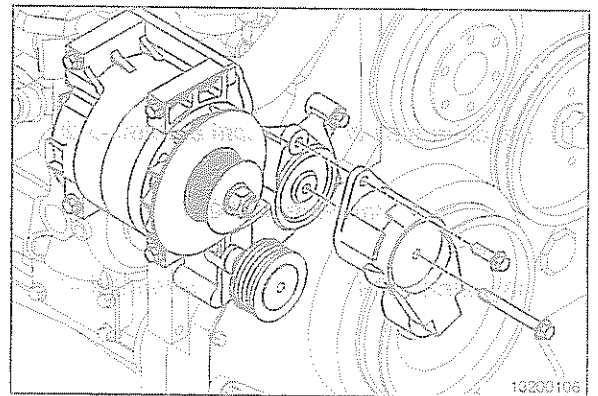
If the tensioner is hitting either of the limits during operation, check the mounting brackets and the belt length. Loose brackets, bracket failure, alternator movement, incorrect belt length, or belt failure can cause the tensioner to hit the limits.

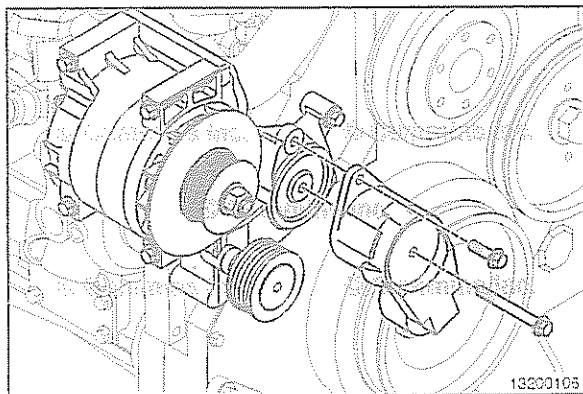
NOTE: If the engine is equipped with a secondary alternator, reference to the vehicle manufacturer for the secondary alternator belt adjustment procedure.

Belt Tensioner, Automatic (Alternator) General Information

A new automatic tensioning single belt fan and alternator drive is the standard option for the QSM industrial engine. The design uses an eight-rib poly-vee belt with an automatic tensioner and a fixed idler.

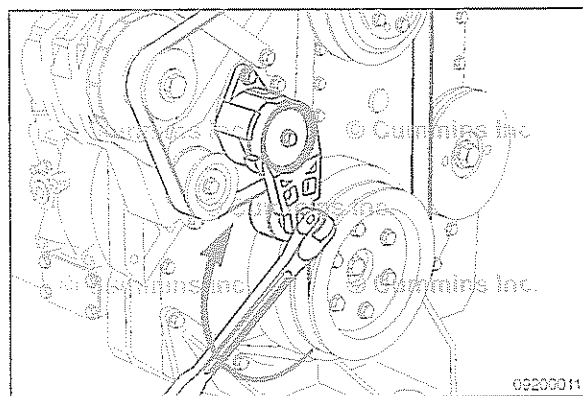
The belt tensioner is designed to operate within the limit of arm movement provided by the cast stops when the belt length and geometry are correct. If the tensioner is hitting either of the limits during operation, check the mounting brackets and the belt length. Loose brackets, bracket failure, alternator movement, incorrect belt length, or belt failure can cause the tensioner to hit the limits.





Remove

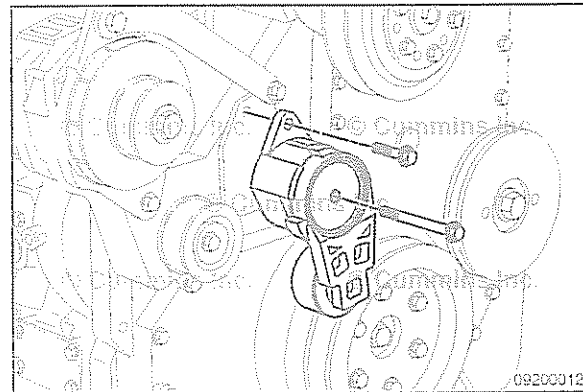
Remove the mounting capscrews and the belt tensioner from the bracket.



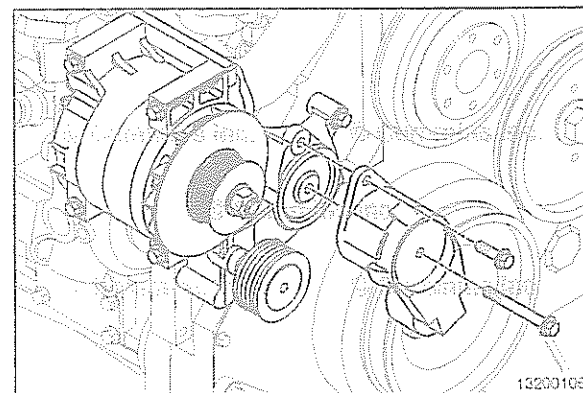
Use a 3/4-inch drive breaker bar to hold back the tensioner and relieve the belt tension.

Rotate the tensioner away from the belt until it stops.

Remove the alternator belt while holding the tensioner back.



Remove the belt tensioner from the engine.



Install

Install the belt tensioner.

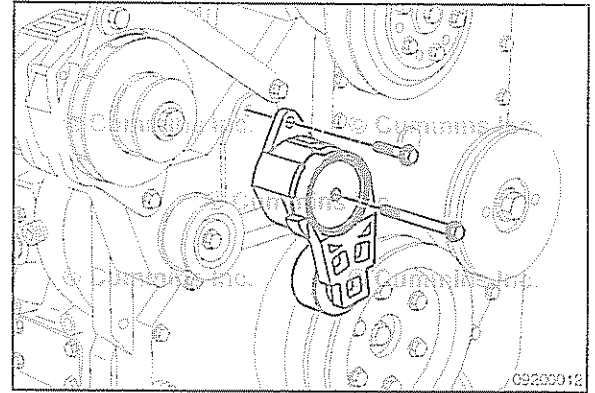


Install and tighten capscrews.

Torque Value: 47 N•m [35 ft-lb]

Tighten the tensioner mounting capscrew.

Torque Value: 47 N•m [35 ft-lb]

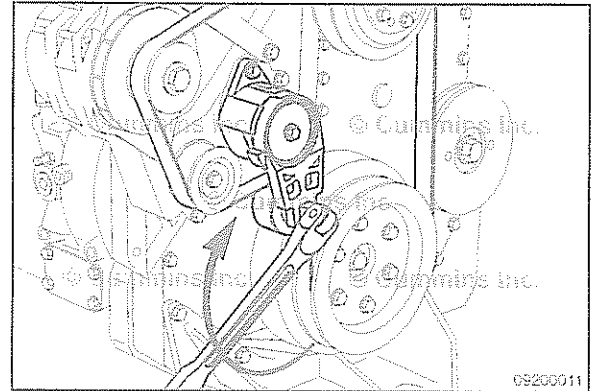


Install the alternator belt over the pulleys while holding the tensioner back with a 3/4-inch drive breaker bar. Use care in working belt over the edge of flanged pulleys.



Release the tensioner and remove the breaker bar.

The belt tensioner is designed to operate well within the limit of arm movement, provided by the cast stops, when the belt length and geometry are correct.



Water Pump

Remove

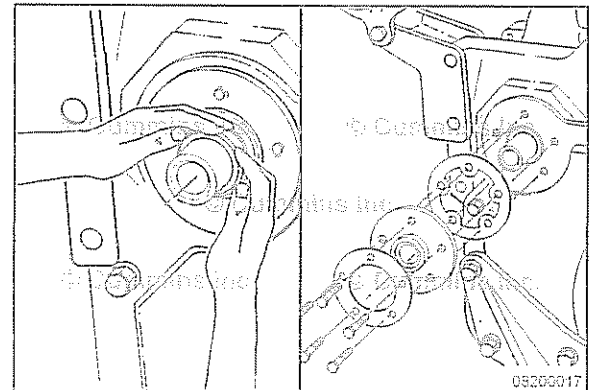
With EGR

Remove the dust seal.

Remove the five water pump oil seal capscrews, clamping ring, oil seal, and gasket.

Remove the dust seal as the seal carrier is removed; or use a heel bar, or similar tool, to pry the dust seal away from the seal case. Then remove the dust seal by hand.

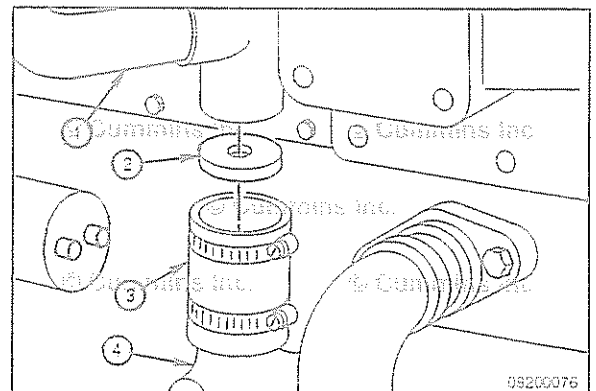
Discard the oil seal and dust seal.

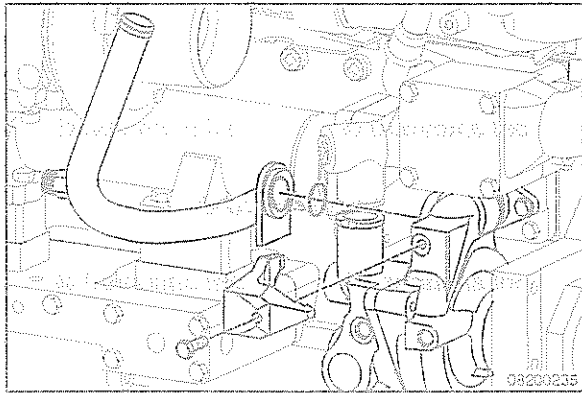


The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

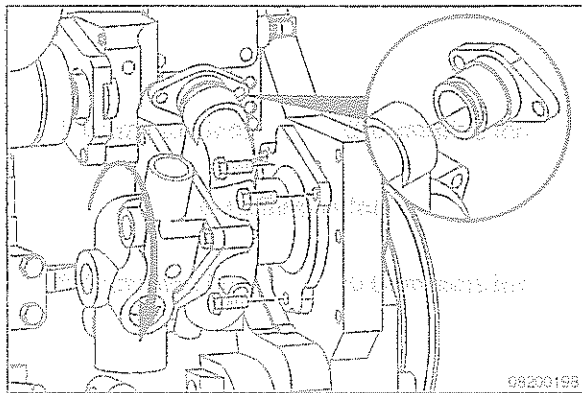
1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.





Loosen the EGR cooler coolant supply lower tube retaining clamp capscrew. Remove the EGR cooler coolant supply lower tube retaining clamp and tube.

Remove and discard the o-ring.

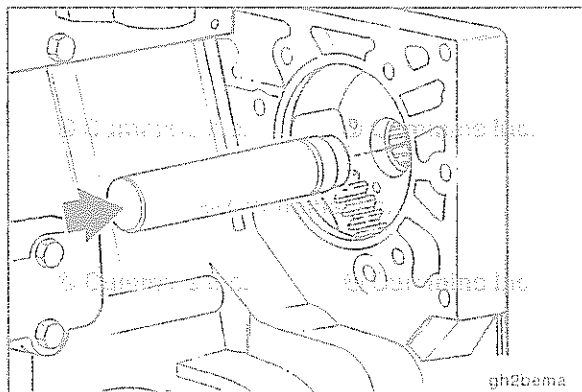


Remove the three water pump mounting capscrews.

Remove the two water pump water transfer connection capscrews and push the water pump transfer connection into the water pump as far as possible to facilitate water pump removal.

Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward as it is being removed to allow the pump to pass the thermostat housing support.

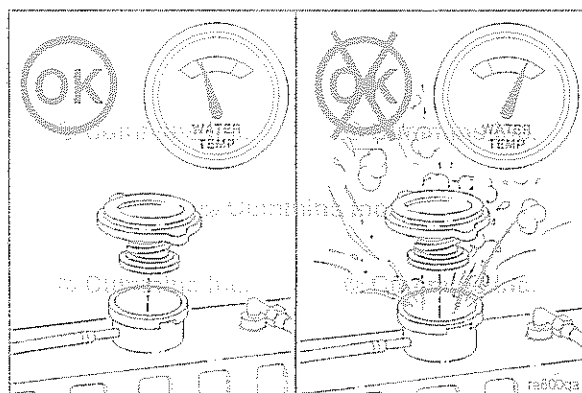
Remove the water pump water transfer connection from the water pump.



Check the needle bearing for damage and freedom of needle rotation. Replace the bearing if it is damaged or if it does not turn freely.



If the bearing is replaced, use bearing installation tool, Part Number 3824117, along with a cup driver to remove the needle bearing from the gear housing. Gently tap the bearing out from the rear side of the housing.



▲ WARNING ▲

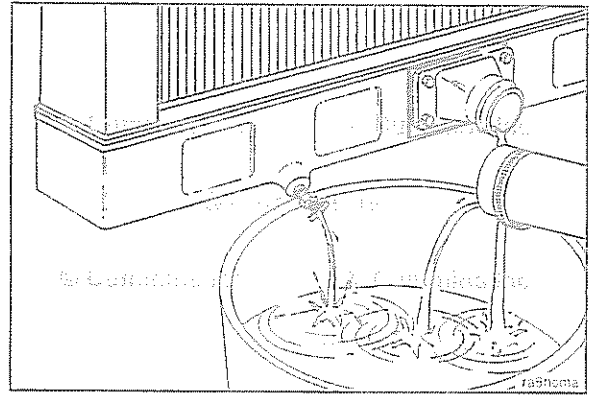
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the pressure cap when the engine is cool.



Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 7.

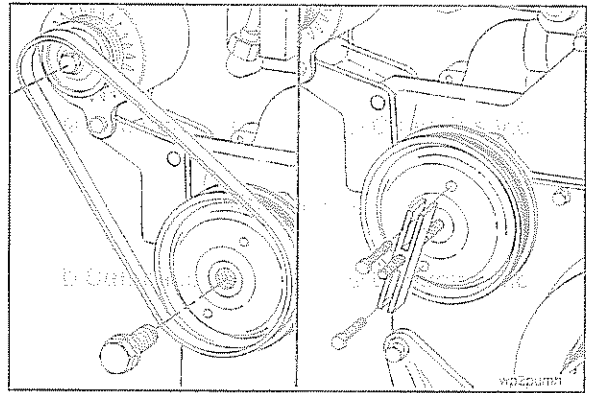


NOTE: Be sure the puller capscrews are threaded all the way through the puller before applying pressure to the puller screw.

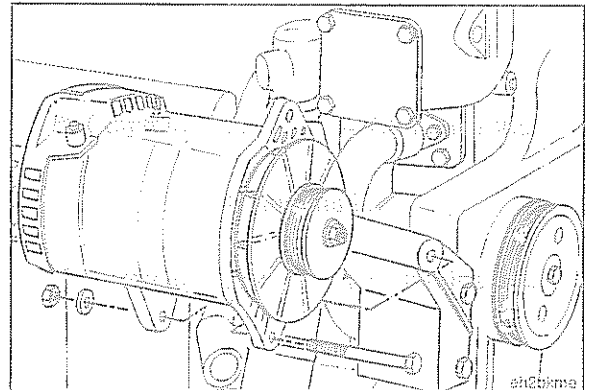
Remove the alternator drive belt. Refer to Procedure 013-005 (Drive Belt, Alternator) in Section A.

Remove the water pump pulley retaining capscrew.

Use the standard pulley puller, Part Number ST-647, or equivalent, and two 5/16 x 18 x 2 capscrews to remove the pulley.



Remove the alternator. Refer to a Cummins Authorized Repair Location.

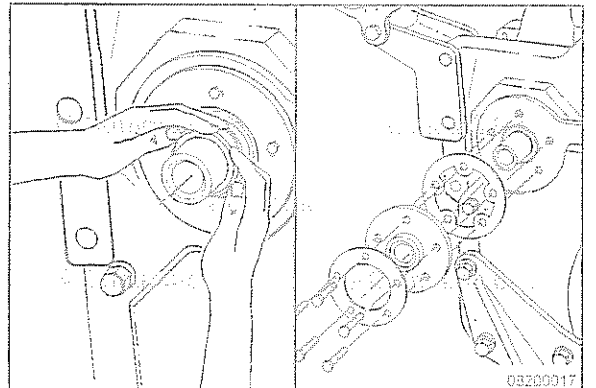


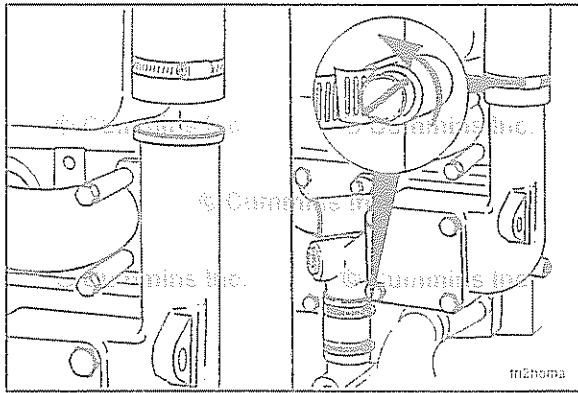
NOTE: Remove the dust seal as the seal carrier is removed, or use a heel bar, or similar tool, to pry the dust seal away from the seal case. Then remove the dust seal by hand.

Remove the dust seal.

Remove the five water pump oil seal capscrews, clamping ring, oil seal, and gasket.

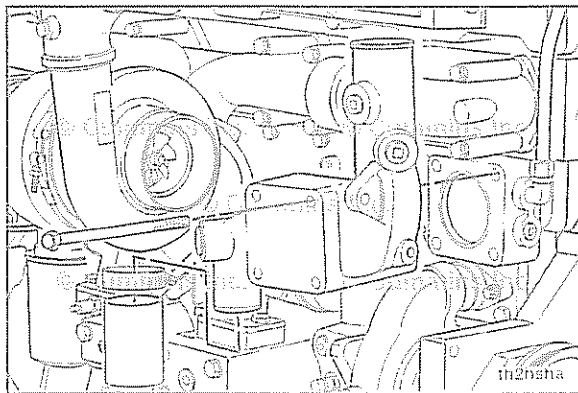
Discard the oil seal and dust seal.



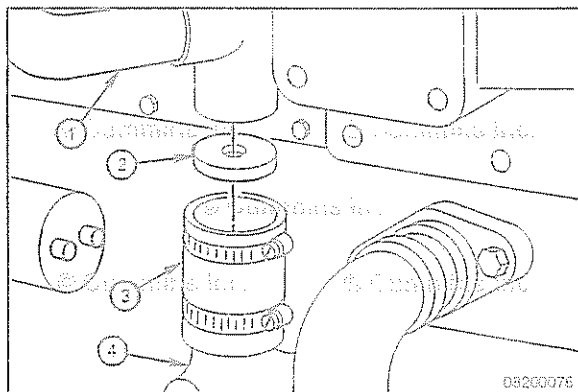


Loosen the coolant bypass hose clamps on both the upper and lower hoses.

Remove the upper coolant hose from the thermostat housing.



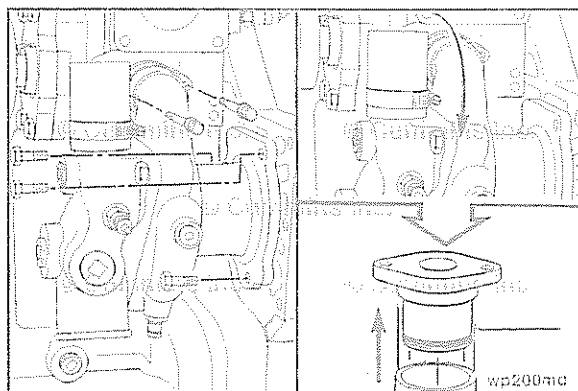
Remove the four thermostat housing mounting capscrews and the thermostat housing.



The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

1. Torque converter coolant supply.
2. Torque converter disc (orifice).
3. Bypass hose.
4. Water pump.



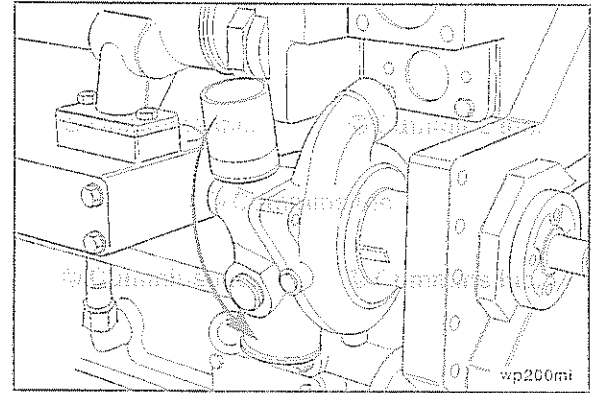
Remove the two water pump water transfer connection capscrews.

Remove the three water pump mounting capscrews.

Rotate the water pump outward so the water transfer connection can be removed from the water pump.

Remove the water transfer connection from the water pump.

Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward as it is being removed to allow the pump to pass the thermostat housing support.



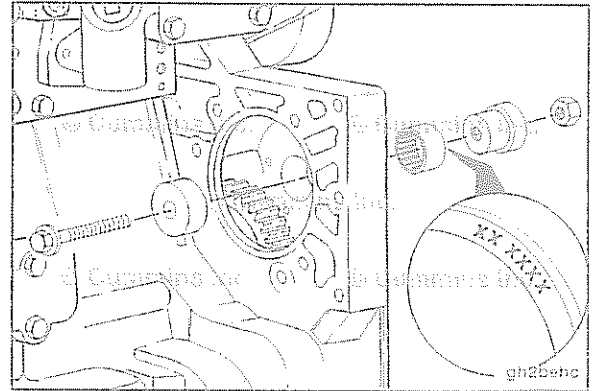
Install

With EGR

If the needle bearing was removed, use bearing installation tool, Part Number 3824117, to install a new needle bearing in the gear housing.

The bearing must be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.

Install the bearing from the front side of the gear housing until the bearing is flush with the front edge of the housing bore.



CAUTION

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

Install the water pump cover. Use a new o-ring seal.

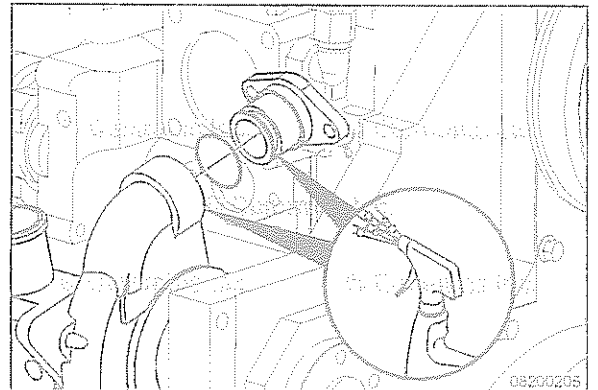
Torque Value: 47 N•m [35 ft-lb]

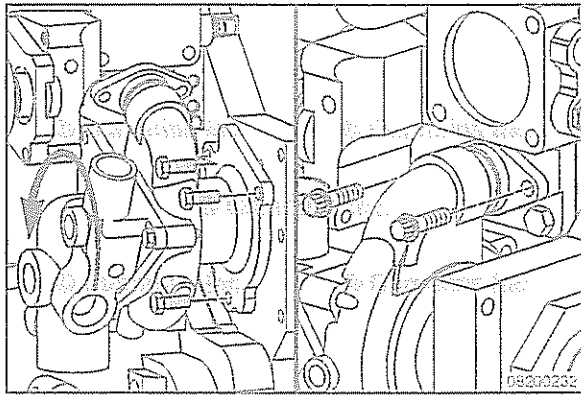
Install the water pump water transfer connection into the water pump.

Install a new o-ring on the water pump mounting flange.

The water pump must be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.

Install the water pump.





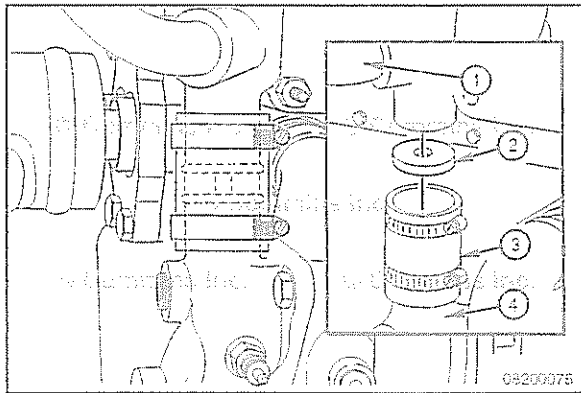
Twist the water pump inward and install the three water pump mounting capscrews.

Torque Value: 47 N•m [35 ft-lb]



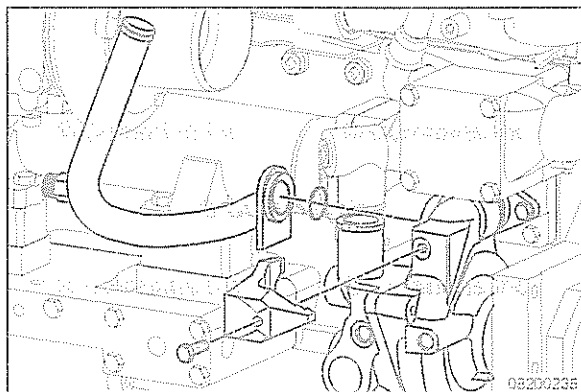
Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 N•m [18 ft-lb]



If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.



CAUTION

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

Install a new o-ring on the EGR cooler coolant supply lower tube.

Insert the EGR cooler coolant supply lower tube into the water pump body.

Install the EGR cooler coolant supply lower tube is seated completely in the water pump body before tightening the EGR cooler coolant supply lower tube retaining clamp capscrew to reduce the possibility of pinching the o-ring.

Tighten the capscrew.

Torque Value: 34 N•m [25 ft-lb]

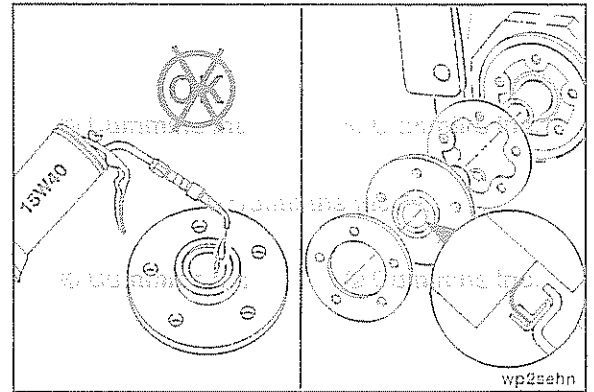
The oil seal must be installed with the lip of the seal and the shaft clean and dry. Do not lubricate. The yellow dust lip must be facing out.



Install the new gasket and oil seal. Use the installation sleeve provided with the new seal to install the seal.

The capscrew threads must be coated with thread sealant, Part Number 3823494, or equivalent, to prevent oil leakage.

Torque Value: Step1	7 N•m	[60 in-lb]
Step2	20 N•m	[180 in-lb]

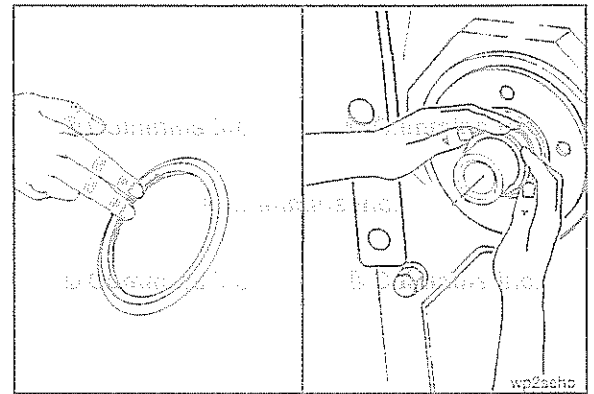


Place a light film of oil or antifreeze on the inside of the oil seal dust seal.



Install the dust seal onto the shaft with a larger outside surface facing the engine.

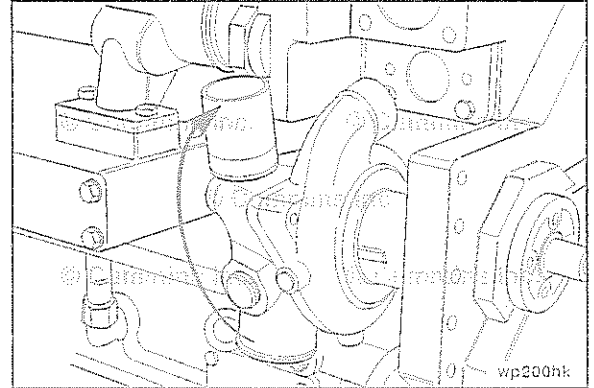
Push the dust seal back by hand on the shaft until the entire dust seal contacts the oil seal case.



NOTE: The water pump must be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.



Install a new o-ring on the water pump mounting flange.
Install the water pump.



Install a new o-ring on the water pump water transfer tube.
Install the connection into the water pump.



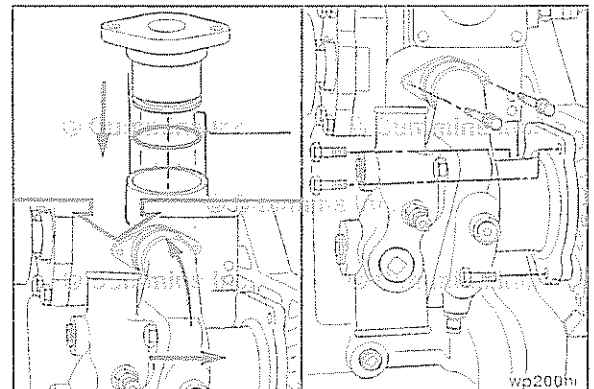
Twist the water pump inward, and install the three water pump mounting capscrews.

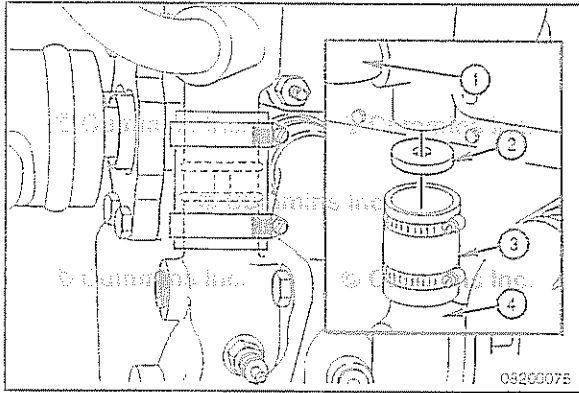


Torque Value: 47 N•m [35 ft-lb]

Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

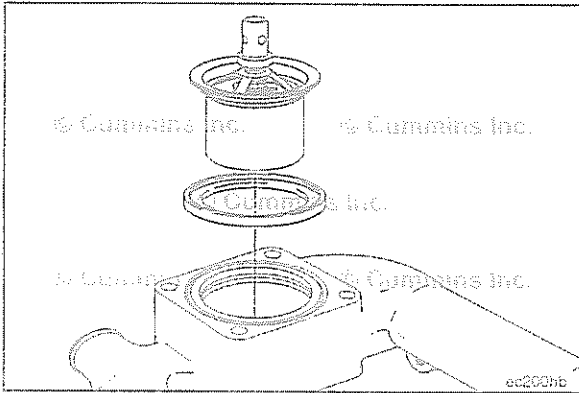
Torque Value: 25 N•m [221 in-lb]





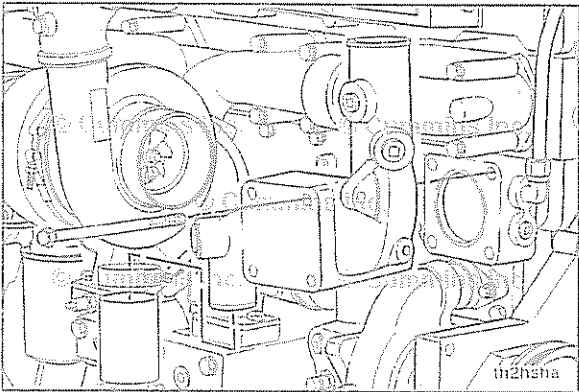
If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter cooler coolant supply.
2. Torque converter cooler disc (orifice).
3. Bypass hose.
4. Water pump.



Install the thermostat in the housing.

Install a new seal in the groove on the thermostat housing mounting surface.

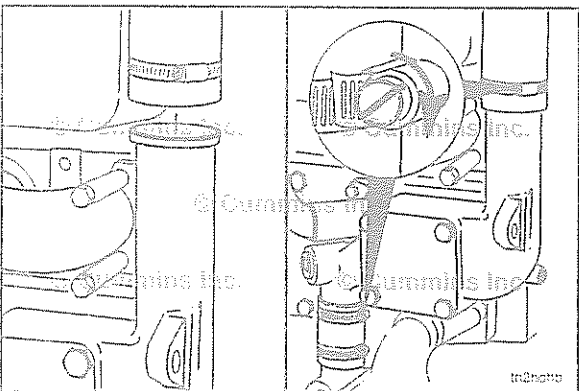


Install the hose on the thermostat housing bypass outlet.

Install the thermostat housing and four mounting capscrews.



Torque Value: 54 N•m [40 ft-lb]



Equally space the bypass hose over the water pump connection and thermostat housing connection, and tighten the bypass hose clamps.



Torque Value: 3 N•m [27 in-lb]

Install the upper and lower coolant hoses. Refer to the OEM's specifications for the correct torque value.

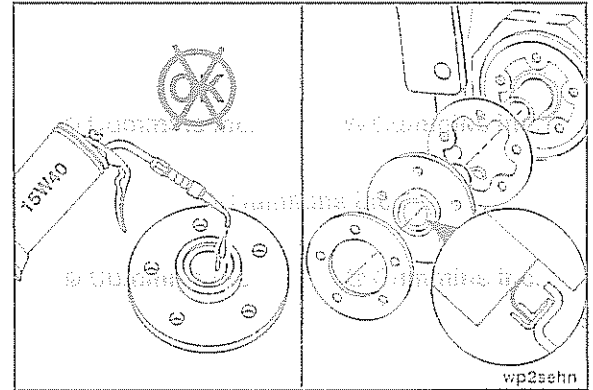


The oil seal must be installed with the lip of the seal and the shaft clean and dry. Do not lubricate. The yellow dust lip must be facing out.

Install the new gasket and oil seal. Use the installation sleeve provided with the new seal to install the seal.

The capscrew threads must be coated with thread sealant, Part No. 3823494, or equivalent, to prevent oil leakage.

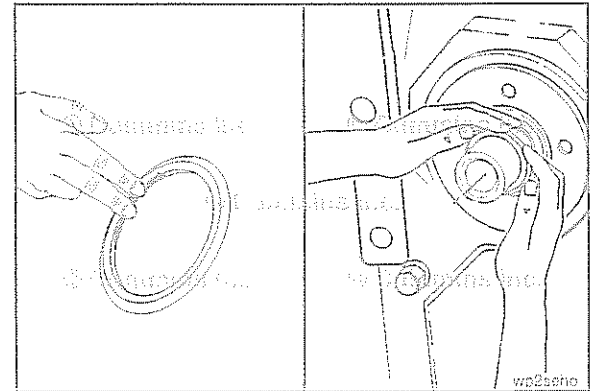
Torque Value: Step1	7 N•m	[62 in-lb]
Step2	20 N•m	[177 in-lb]



Place a light film of oil or antifreeze on the inside diameter of the dust seal.

Install the dust seal onto the shaft with the larger outside diameter facing the engine.

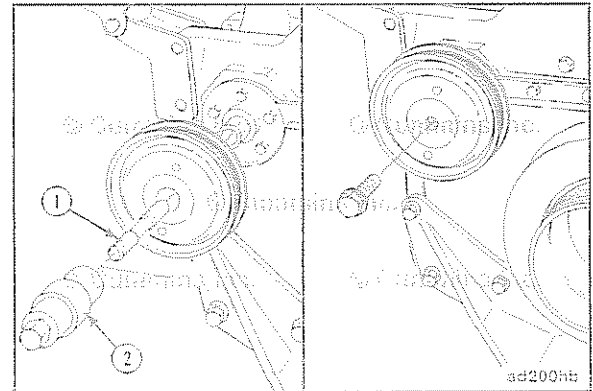
Push the dust seal back by hand on the shaft until the entire dust seal contacts the oil seal case.



Use pulley pusher adapter (1), Part Number 3377401, or equivalent, and pulley pusher (2), Part Number 3376326, or equivalent, to install the pulley.

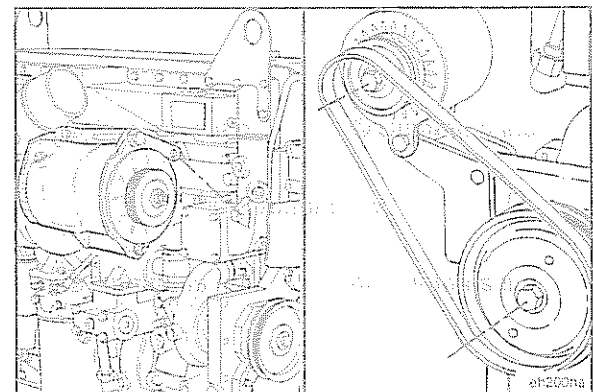
Install the capscrew in the shaft.

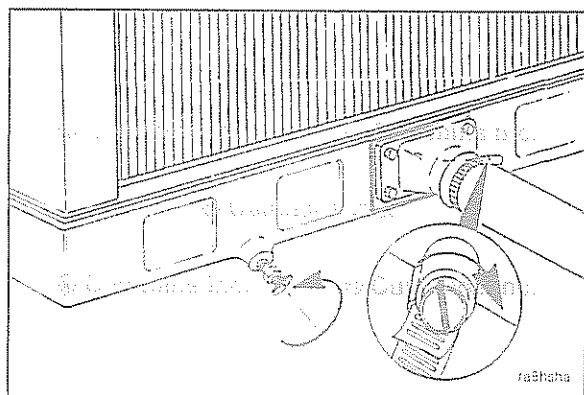
Torque Value: 75 N•m [55 ft-lb]



Install the alternator. Refer to a Cummins Authorized Repair Location.

Install and adjust the alternator drive belt. Refer to 013-005 (Drive Belt, Alternator) in Section A.

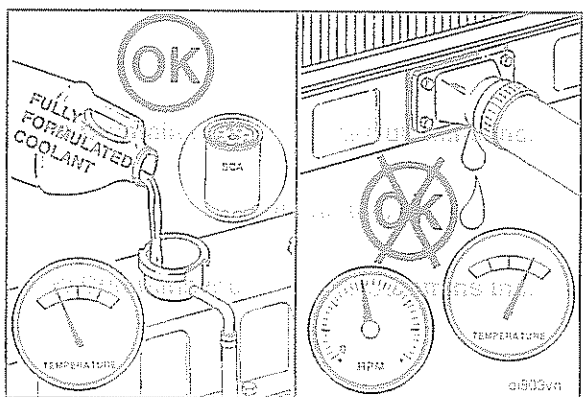




Close the cooling system draincock, and install the lower coolant hose.



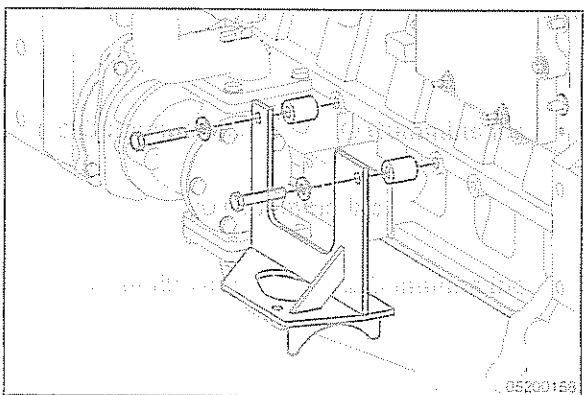
Tighten the hose clamp(s). Refer to the manufacturer's specifications for the correct torque value.



The correct concentration of coolant additives must be used in the cooling system. Refer to Procedure 018-004 (Coolant Recommendation and Specifications) in Section V.

Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 7.

Operate the engine until it reaches a temperature of 71° C [160°F] and check for coolant leaks.



Sea Water Pump Preparatory Steps

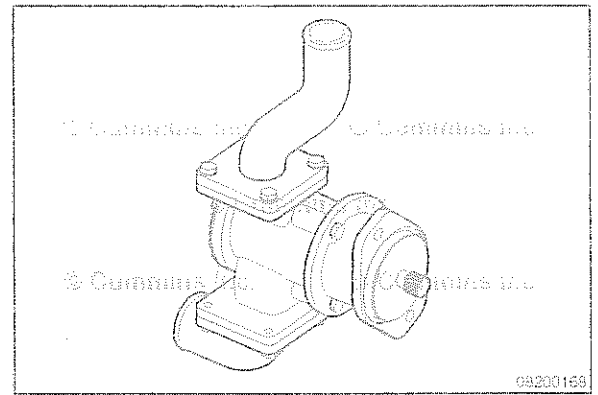
⚠CAUTION⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations

- Shut off the sea water inlet valves.
- Shut off the fuel supply and return valves (if a major kit repair is planned).
- Remove the fuel cooler (if a major kit repair is planned).
- Remove the fuel cooler mounting bracket for access to the sea water pump cover.

Remove

If a major kit repair is being performed, remove the sea water pump.



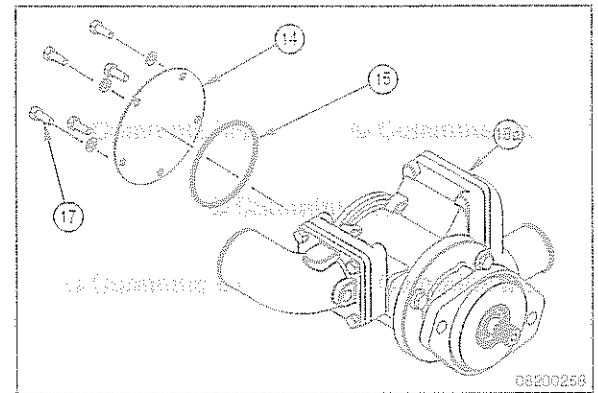
Disassemble

Marine Applications

If a minor kit repair is being performed, remove the cover plate.

Remove the five capscrews (17) and the washers and the cover (14) from the sea water pump.

Remove and discard the quad-ring (15).



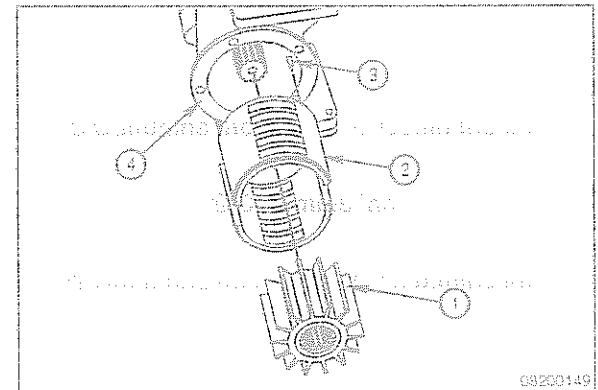
⚠CAUTION⚠

Do not pry against the cam liner to remove the impeller as this will break the cam liner.

NOTE: The two-piece cam liner will separate when removed. Do not drop the cam liner.

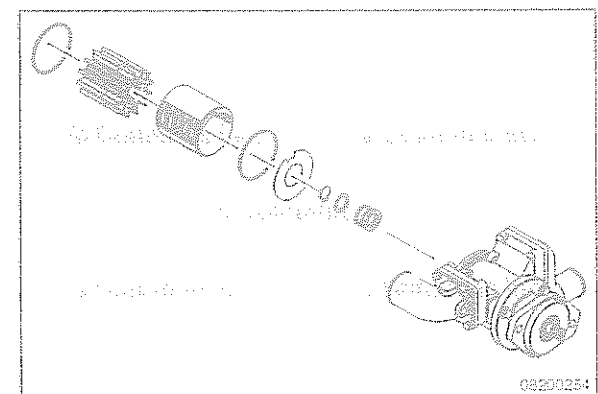
Check the direction of the impeller blades to make sure the impeller is properly installed at assembly.

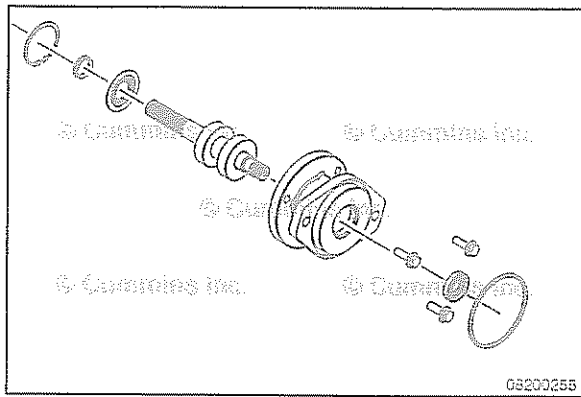
Remove the impeller (1), two-piece cam liner (2), and pin (3) from the sea water pump (4).



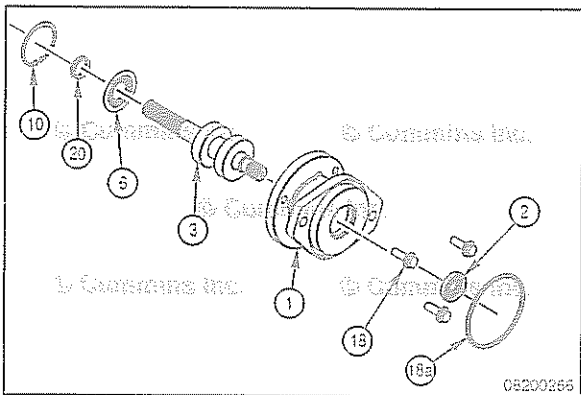
Remove the second quad ring and wearplate.

Remove the retaining ring from the shaft. Slide the washer and mechanical seal off of the shaft.





- Remove the bolts holding the body to the housing.
- Slide the housing off of the shaft and body assembly.
- Press the seal seat out of the housing.
- Remove the retaining ring from the body.
- Press the bearing and shaft assembly out of the body.
- Remove V-ring and bearing shield from the shaft.
- Remove the lip seal.



Assemble

Marine Applications

- Press the lip seal (2) into the pump body (1) with the backing ring of the lip seal toward the main pump housing.
- Align the lip seal so that it is uniformly pressed in the pump body. Press the shaft and bearing assembly (3) into the pump body (1).
- Slide the bearing shield (6) into the pump body and against the inner bearing.
- Place the V-ring (20) on the shaft against the bearing shield, with the open end of the V-ring toward the bearing assembly.
- Use snap ring pliers to install the retaining ring (10) into the body.



Do not use petroleum based lubricant. A petroleum based lubricant will cause damage to the impeller.

NOTE: Grease can be used to hold the pin in place during installation.

Press the mechanical seal (27) into the housing with the white ceramic seal face toward the impeller.

Lubricate the outside of the seat boot with a water-soluble lubricant or soapy water for easier assembly.

Slide the housing (22) over the shaft and body assembly.

Install the three bolts (18).

Torque Value: 34 N•m [25 ft-lb]

Install the washer (12) onto the shaft and use snap ring pliers to install the retaining ring (11) into the groove in the shaft (3) to compress the mechanical seal.

Install the wearplate (13) with the hole in the plate lined up with the slot in the housing.

Install the quad ring (15) into the housing on top of the wearplate.

Assemble the cam halves (9) together and install the pin (19) into the cam liner.

Place the cam assembly into the housing, align the pin with the slot in the housing.

Insert the impeller (8) in the housing (22) using a twisting motion.

Make sure the impeller blades are in the same orientation as when removed.

Apply a silicon lubricant to aid in the installation of the impeller.

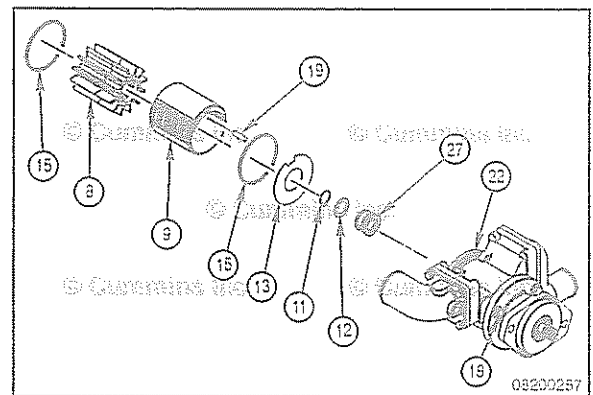
Insert the remaining quad ring (15) into the groove.

Install the cover (14) onto the housing (18a).

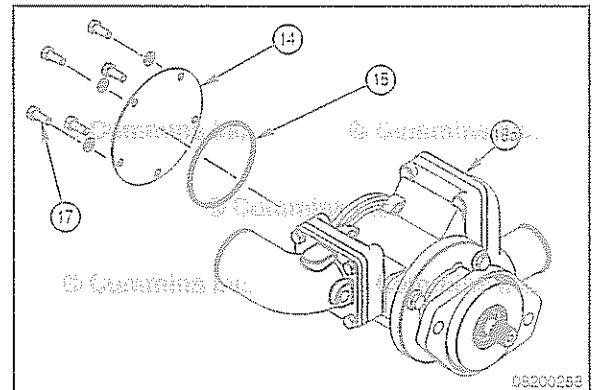
Install bolts (17).

Torque Value: 11 N•m [96 in-lb]

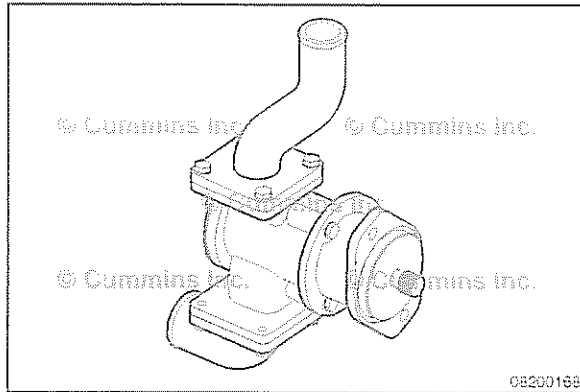
Install the front flange sealing o-ring onto the pump body.



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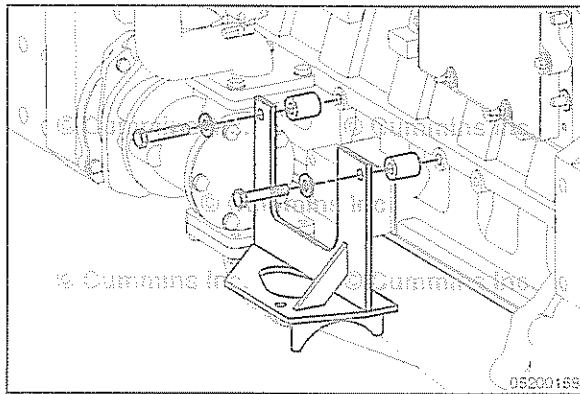


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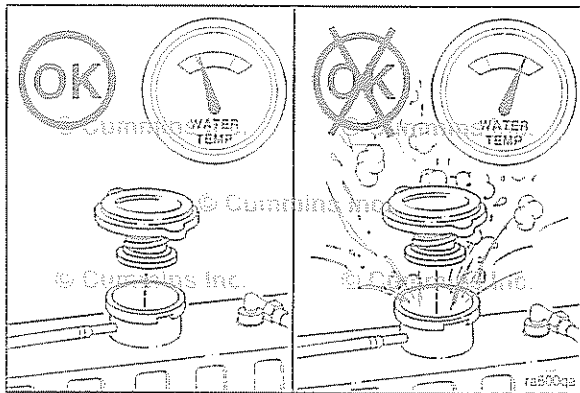
Install

If a major kit repair was performed, install the sea water pump.



Finishing Steps

- Install the fuel cooler mounting bracket.
- Install the fuel cooler (if a major repair kit was installed).
- Open the fuel supply and return valves (if the fuel cooler was removed).
- Open the sea water inlet valve.
- Operate the engine and check for leaks.

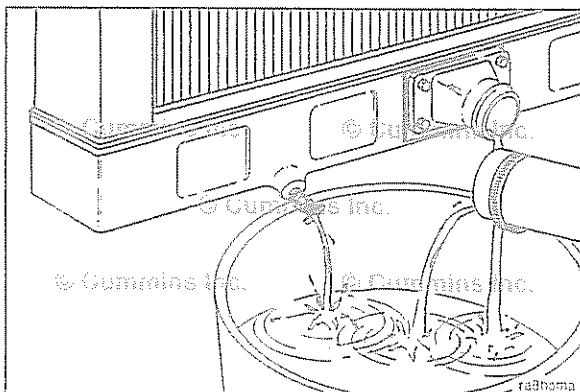


Coolant Thermostat Remove

▲ WARNING ▲

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the coolant system pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the pressure cap when the engine is cool.



▲ WARNING ▲

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

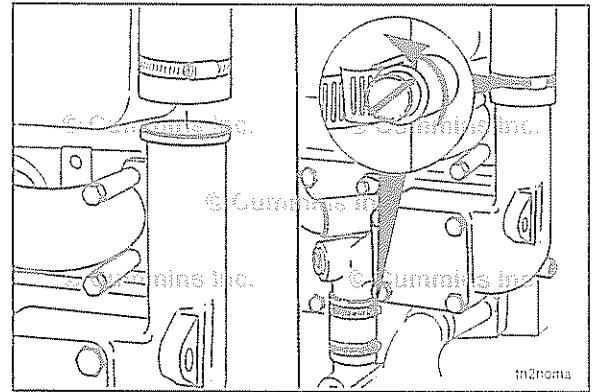
Drain the cooling system as follows:

- Open the radiator draincock.
- Remove the lower radiator hose.

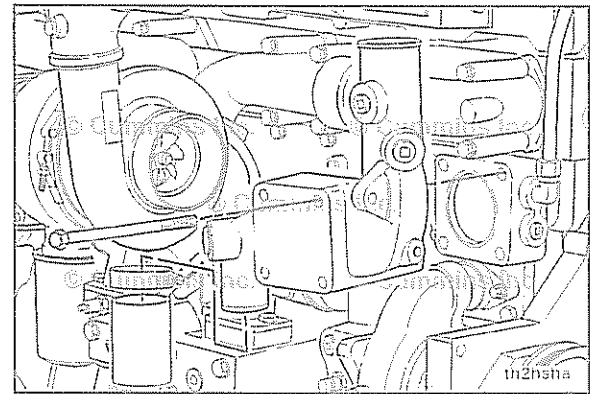
Remove the upper coolant hose from the thermostat housing.

Loosen the coolant bypass hose clamps.

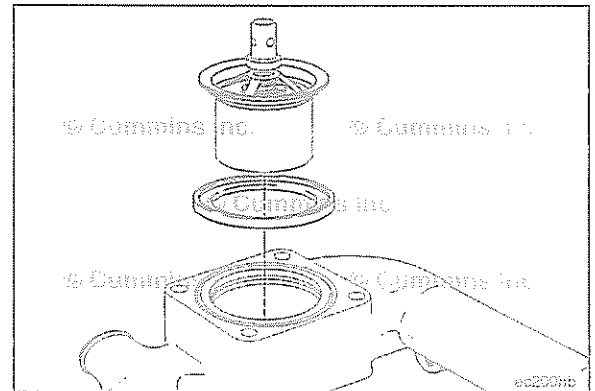
NOTE: Some models could have a converter cooler disc located in the bypass hose.



Remove the four thermostat housing mounting capscrews and the thermostat housing.



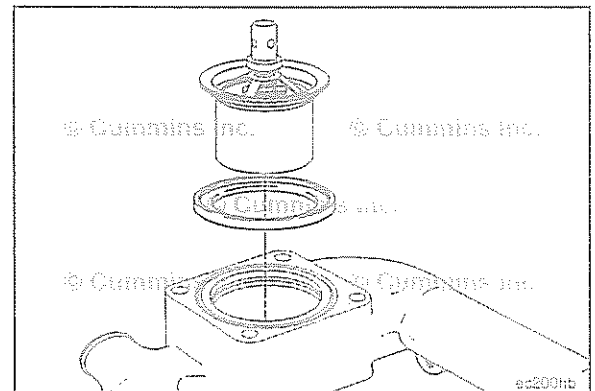
Remove the thermostat from the housing.

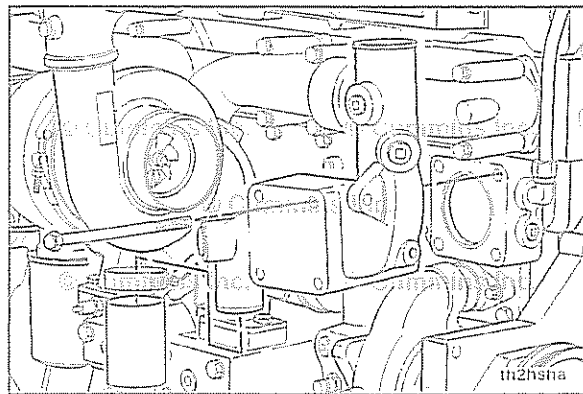


Install

Install the thermostat in the housing.

Install a new o-ring seal in the groove on the thermostat housing mounting surface.



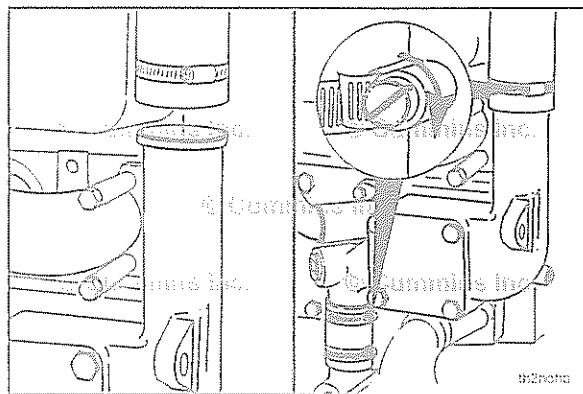


Install the hose on the thermostat housing bypass outlet.
Install the thermostat housing and four mounting capscrews.



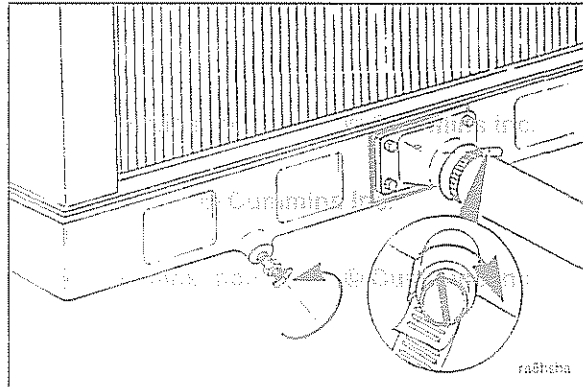
Tighten the mounting capscrews.

Torque Value: 54 N•m [40 ft-lb]



Install the upper coolant hose to the thermostat housing outlet.

Torque Value: 3 N•m [30 in-lb]

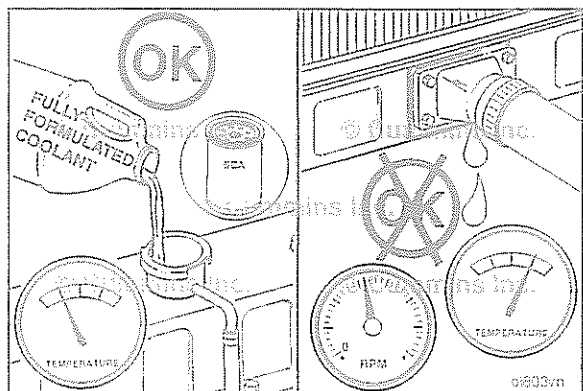


Close the cooling system draincock, and install the lower coolant hose.

Tighten the hose clamp.



Refer to the OEM's specifications for the correct torque value.



The correct concentration of coolant additives must be used in the cooling system. Refer to Procedure 018-003 (Cummins Recommendations and Specifications) in Section V.



Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 7.

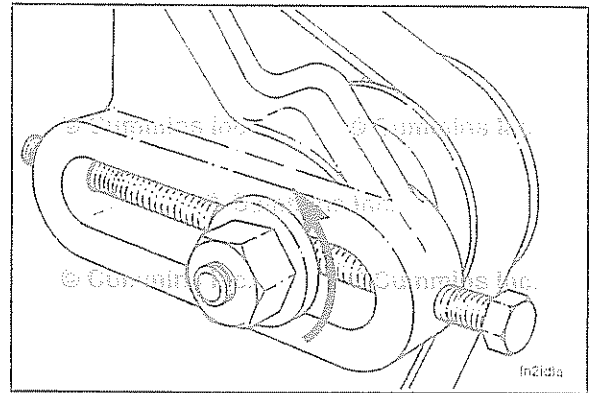
Operate the engine until it reaches 80°C [180°F], and check for coolant leaks.

Fan Drive Idler Pulley Assembly

Remove

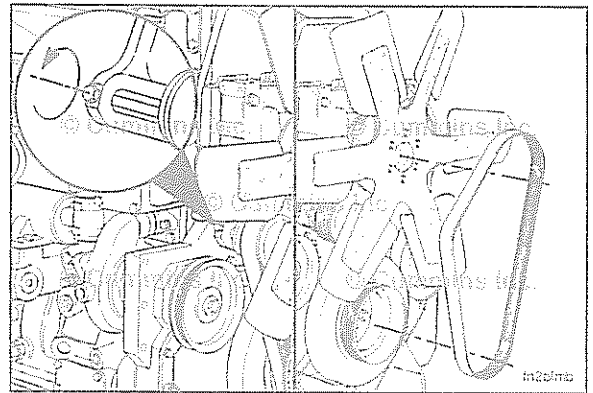
NOTE: For non-Cummins fan hub installations, contact the OEM.

Loosen the fan idler pulley shaft locknut.



Loosen the adjusting mechanism and move the fan idler pulley and fan pulley centers as close as possible.

Remove the fan belt.

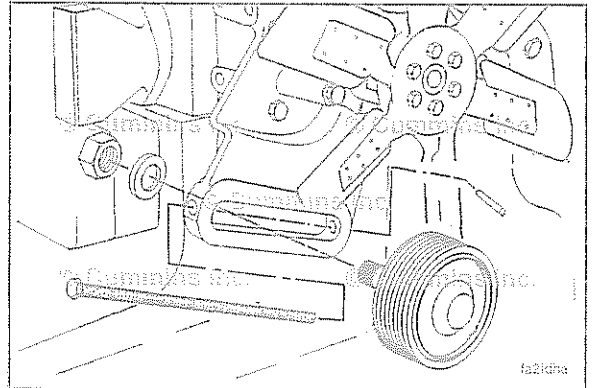


Remove the roll pin and washer from the idler pulley adjusting screw.

Remove the locknut and washer from the back of the idler pulley shaft.

Remove the adjusting screw.

Remove the fan idler pulley from the fan hub support bracket.

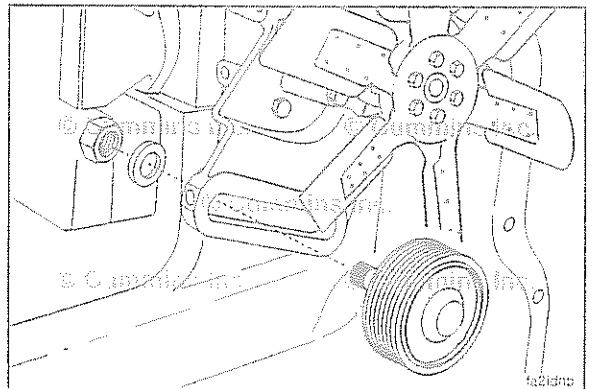


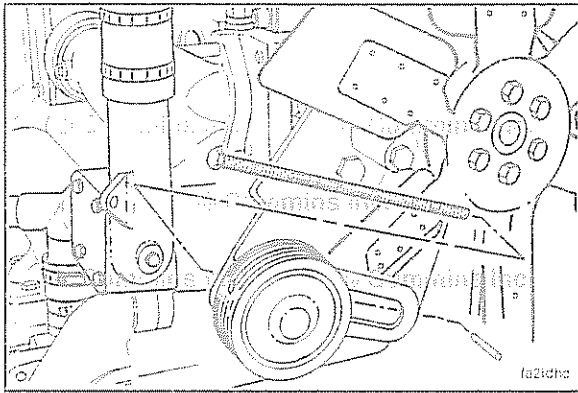
Install

NOTE: Do not tighten the locknut until the fan drive belt has been installed and adjusted.

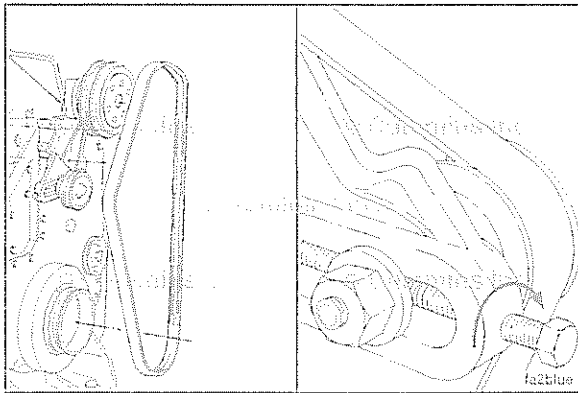
Install the idler pulley in the fan hub support bracket.

Install the washer and locknut on the idler pulley shaft.





Install the adjusting screw in the idler pulley shaft.
Turn the adjusting screw in far enough to install the washer and roll pin in the shaft at the bottom of the fan hub support bracket.



Install and adjust the fan drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section A.



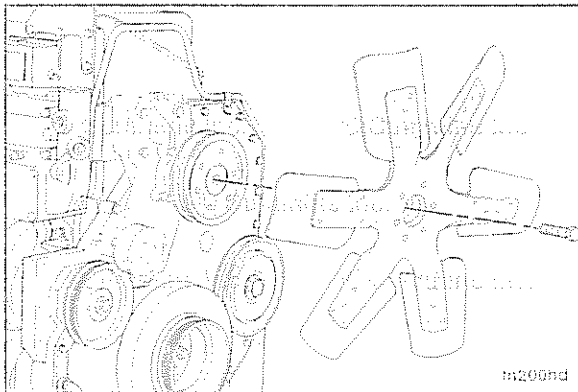
Refer to Procedure 018-005 (Drive Belt Tension) in Section V for proper belt tension value.



Tighten the idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]

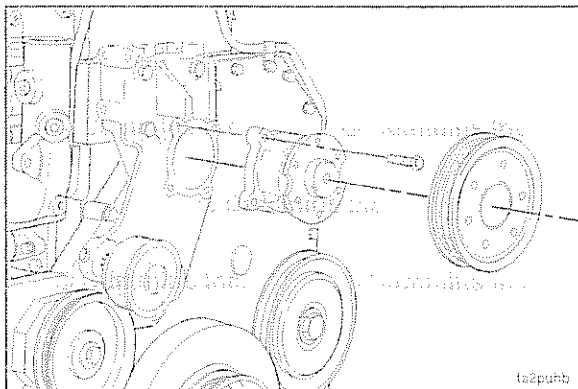
Check the belt tension again after the locknut is tightened.



Fan Hub, Belt Driven Remove

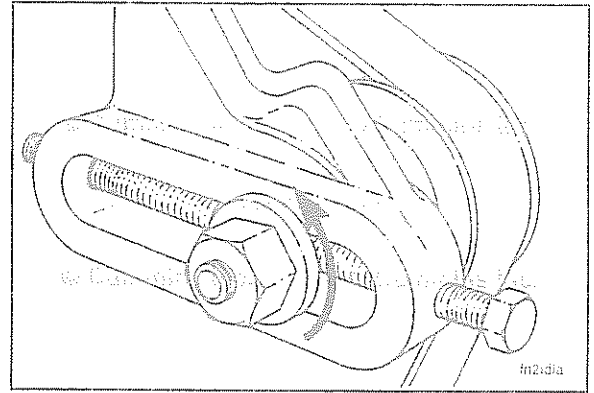


Remove the fan and clutch assembly. Refer to Procedure 008-040.

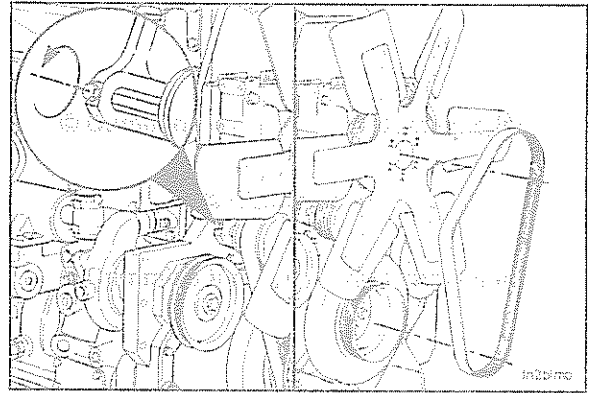


Remove the fan drive pulley.
Remove the four capscrews and the fan hub.

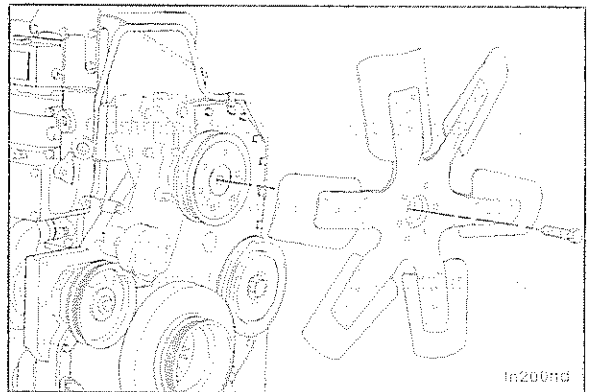
Loosen the fan idler pulley shaft locknut.



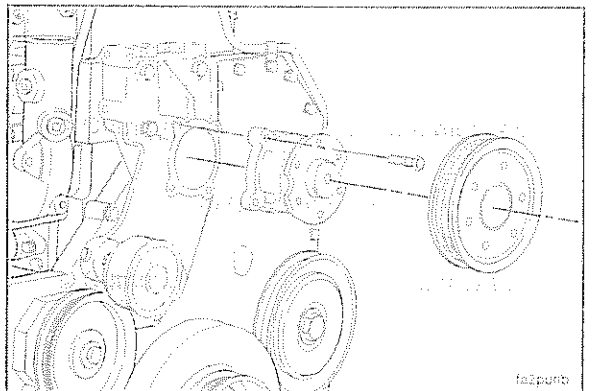
Loosen the adjusting mechanism, and move the fan idler pulley and fan pulley centers as close as possible.
Remove the fan belt.

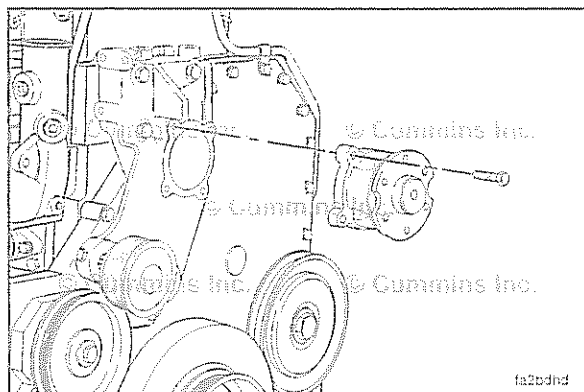


Remove the fan and clutch assembly. Refer to a Cummins Authorized Repair Location.



Remove the fan drive pulley.
Remove the four capscrews and the fan hub.



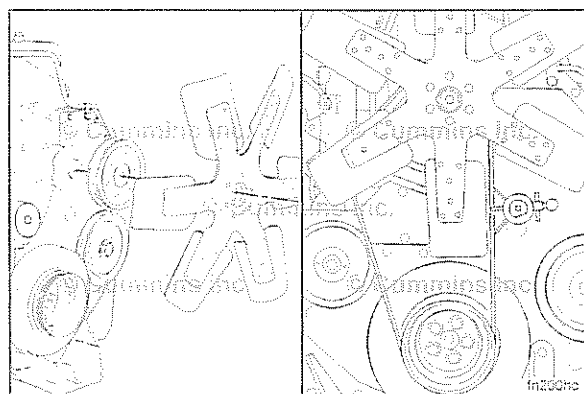


Install

Install the new fan hub and four capscrews.



Torque Value: 47 N•m [35 ft-lb]



Install the fan drive pulley.

Install the fan clutch and assembly. Refer to Procedure 008-040.

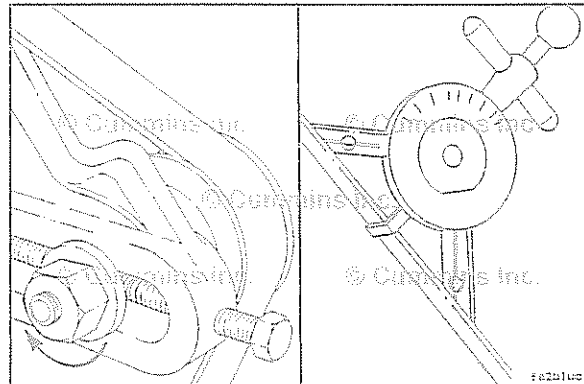


Install and tighten the mounting capscrews.

Torque Value: 68 N•m [50 ft-lb]



Install, adjust, and tighten the fan drive belt. Refer to Procedure 008-002.

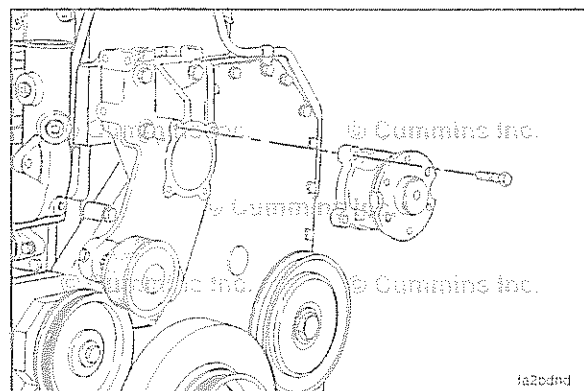


Tighten the fan idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]



Check the belt tension again after the locknut is tightened.



Install the new fan hub and four capscrews.

Torque Value: 47 N•m [35 ft-lb]



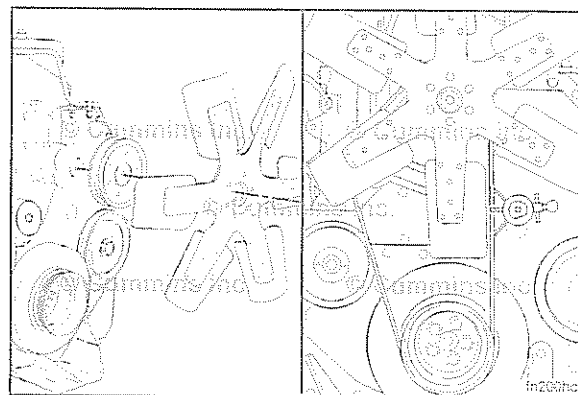
Install the fan drive pulley.

Install the fan and clutch assembly.

Install and tighten the mounting capscrews.

Torque Value: 68 N•m [50 ft-lb]

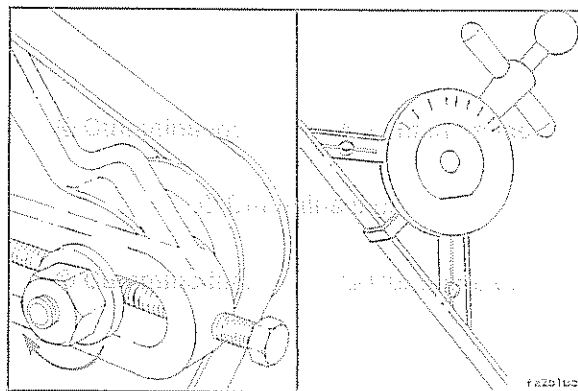
Install, adjust, and tighten the fan drive belt. Refer to Procedure 018-005 (Drive Belt Tension) in Section V for the correct belt tension.



Tighten the fan idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]

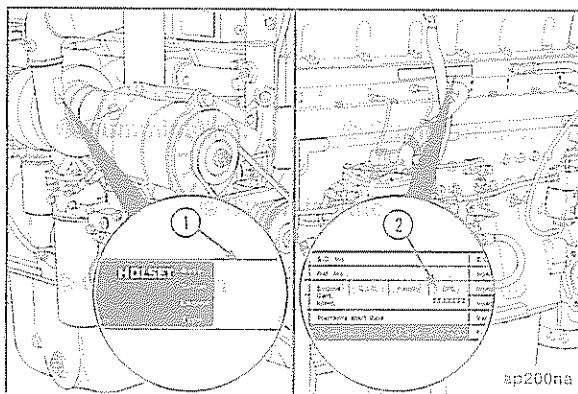
Check the belt tension again after the locknut is tightened.



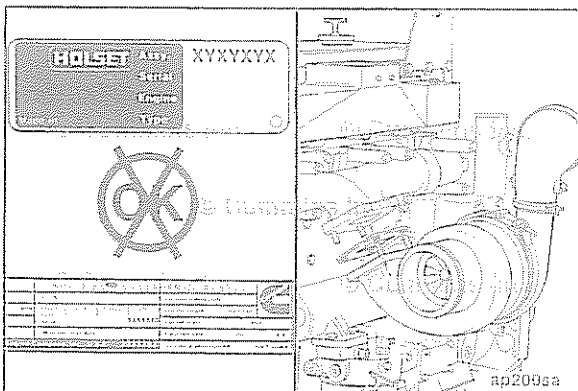
Turbocharger

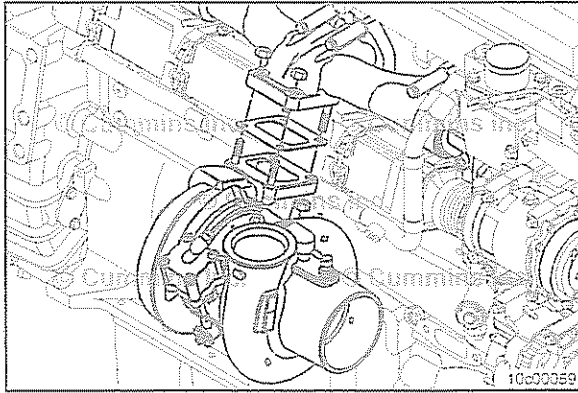
Check for Correct Component

Compare the assembly number (1) on the turbocharger data tag with the turbocharger specified in the engine control parts list (CPL) number (2) listed on the engine data tag.



If the correct turbocharger was not installed, remove it and install the correct turbocharger.





Remove Variable Geometry

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠

When installing the lifting eye, make sure the shoulder of the lifting eye is bottomed against the bearing housing. Failure to do so can result in failure of the lifting eye and personal injury.

⚠ CAUTION ⚠

If the lifting eye is installed in the turbocharger bearing housing, it is to be used exclusively for turbocharger removal and installation. It is not to be used in removal of the exhaust manifold, or engine. Doing so will cause damage the turbocharger.

⚠ CAUTION ⚠

The turbocharger actuator must not be used as a lifting mechanism. Doing so will cause damage to the turbocharger actuator.

NOTE: ISM CM870 - The capscrew in top of the turbocharger bearing housing can be removed and replaced with a lifting eye to aid in removal of the turbocharger.

NOTE: ISM CM876 - Thread a lifting strap around the turbocharger bearing housing. Take care not to bend the coolant lines when lifting the turbocharger.

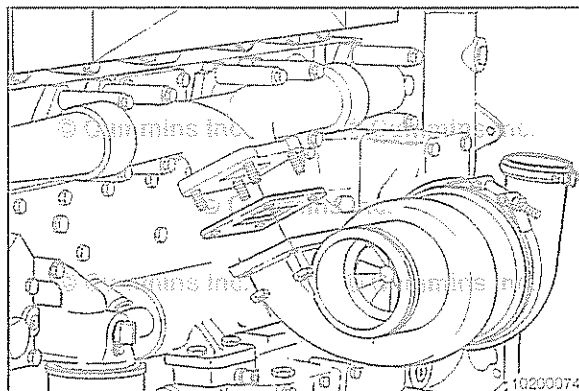
Care must be taken not to damage the turbocharger compressor inlet air temperature sensor, the turbocharger speed sensor, and the turbocharger actuator coolant lines when removing the turbocharger from the engine.

Remove the four turbocharger mounting nuts.

Remove the nuts from the turbocharger studs and remove the turbocharger.

NOTE: If the turbocharger mounting nuts do not loosen freely, split the nuts to avoid breaking a mounting stud.

Remove the turbocharger and discard the gaskets.



Wastegated

Remove the four turbocharger mounting nuts.

Remove the turbocharger and gasket.

Install

Variable Geometry

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠

When installing lifting eye, make sure the shoulder of the lifting eye is bottomed against the bearing housing. Failure to do so can result in failure of lifting eye and personal injury.

⚠ CAUTION ⚠

If the lifting eye is installed in the turbocharger bearing housing, it is to be used exclusively for turbocharger removal and installation. It is not to be used in removal of the exhaust manifold, or engine. Doing so will cause damage to the turbocharger.

⚠ CAUTION ⚠

Do not rotate the turbocharger turbine housing. Loosening the turbine v-band and rotating the turbine housing may cause damage to an internal variable geometry mechanism.

⚠ CAUTION ⚠

The turbocharger actuator must not be used as a lifting mechanism. Doing so will cause damage to the turbocharger actuator.

NOTE: ISM CM870 - The capscrew in the top of the turbocharger bearing housing can be removed and replaced with a lifting eye to aid in removal of the turbocharger.

NOTE: ISM CM876 - Thread a lifting strap around the turbocharger bearing housing. Take care to not bend the coolant lines when lifting the turbocharger.

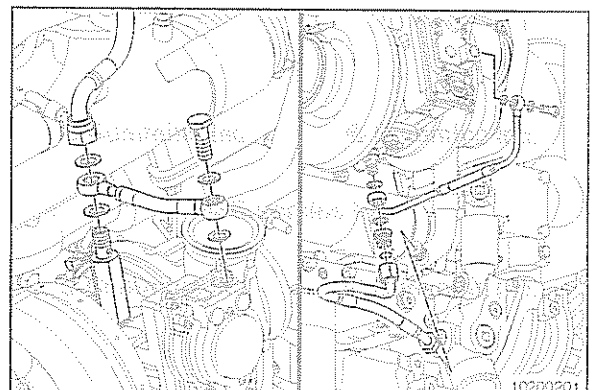
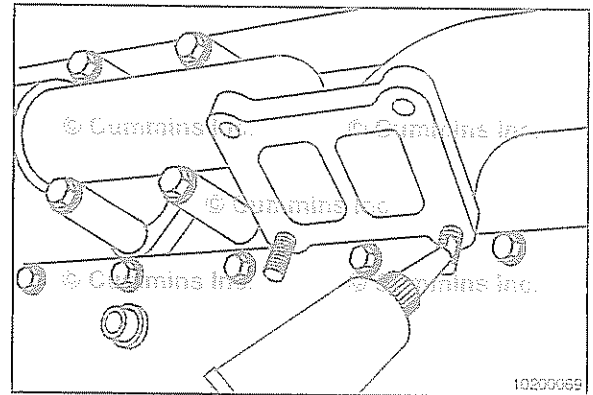
Care must be taken not to damage the turbocharger compressor inlet air temperature sensor, the turbocharger speed sensor and the turbocharger actuator coolant lines when installing the turbocharger.

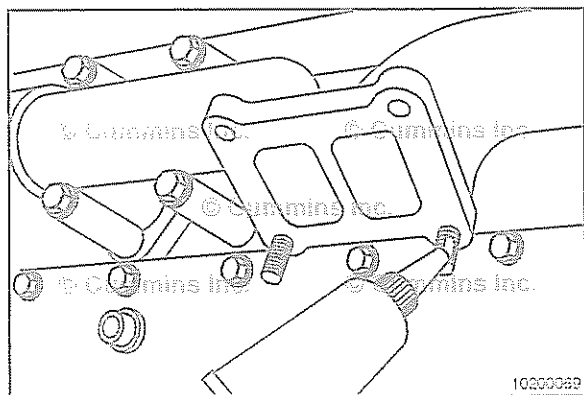
Install a new gasket, the turbocharger, and the four mounting nuts. Tighten the mounting nuts.

Torque Value: 88 N•m [65 ft-lb]

If installing a new turbocharger, install the coolant inlet and outlet fittings. Refer to Procedure 010-041.

If installing a new turbocharger, install the oil supply fitting. Refer to Procedure 010-046.



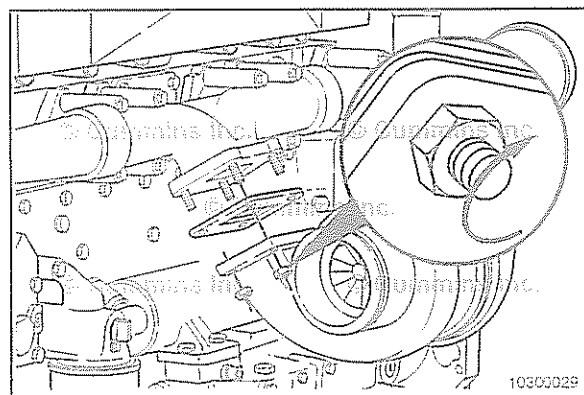


Wastegated

NOTE: For wastegated turbochargers, refer to Procedure 010-050 for wastegate actuator installation.



Apply a film of anti-seize compound, Part Number 3824879, or equivalent, to the turbocharger mounting studs.



Use a new gasket when installing the turbocharger.

Install and tighten the four mounting nuts.

Torque Value: 61 N•m [45 ft-lb]



Air Starting Motor

General Information

The air starting motor system (tanks, line sizes, and valves) is designed and installed by the OEM and starting motor suppliers. Refer any questions about the air starting systems to the manufacturer.

Air Starting Motor Maintenance

- Do not operate the air starting motor with air pressure lower than 480 kPa [70 psi].
- Maintain the air compressor according to the recommendations outlined in the manual.
- For maximum efficiency, the hoses, tubes, and lines must not leak.
- Refer to the OEM and starting motor manufacturer's manuals for specific information regarding the starting motors, valves, and systems.

Refrigerant Compressor Drive

General Information

Industrial Applications

A new refrigerant compressor drive system is offered on the QSM11 industrial. The new drive system has an accessory drive pulley, a six-rib poly-vee belt with automatic tensioning and a backside idler for use with a Sanden rotary compressor.

Install

Variable Geometry



WARNING

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



CAUTION

When installing lifting eye, make sure the shoulder of the lifting eye is bottomed against the bearing housing. Failure to do so can result in failure of lifting eye and personal injury.



CAUTION

If the lifting eye is installed in the turbocharger bearing housing, it is to be used exclusively for turbocharger removal and installation. It is not to be used in removal of the exhaust manifold, or engine. Doing so will cause damage to the turbocharger.



CAUTION

Do not rotate the turbocharger turbine housing. Loosening the turbine v-band and rotating the turbine housing may cause damage to an internal variable geometry mechanism.



CAUTION

The turbocharger actuator must not be used as a lifting mechanism. Doing so will cause damage to the turbocharger actuator.

NOTE: ISM CM870 - The capscrew in the top of the turbocharger bearing housing can be removed and replaced with a lifting eye to aid in removal of the turbocharger.

NOTE: ISM CM876 - Thread a lifting strap around the turbocharger bearing housing. Take care to not bend the coolant lines when lifting the turbocharger.

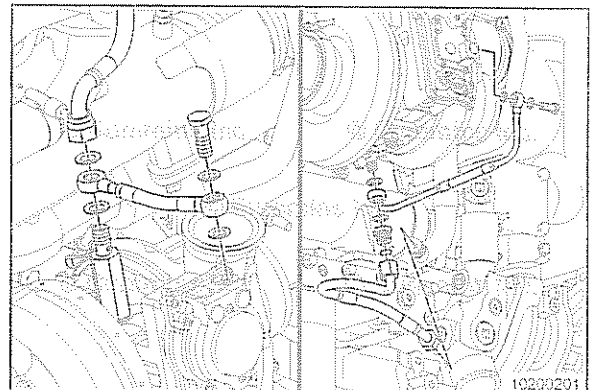
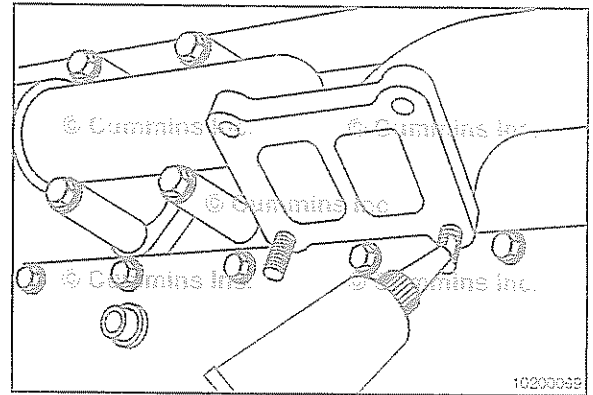
Care must be taken not to damage the turbocharger compressor inlet air temperature sensor, the turbocharger speed sensor and the turbocharger actuator coolant lines when installing the turbocharger.

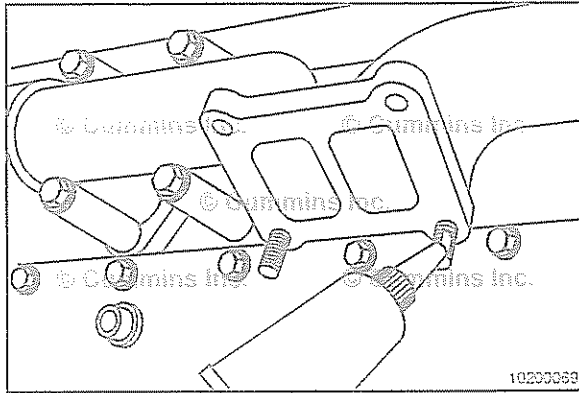
Install a new gasket, the turbocharger, and the four mounting nuts. Tighten the mounting nuts.

Torque Value: 88 N•m [65 ft-lb]

If installing a new turbocharger, install the coolant inlet and outlet fittings. Refer to Procedure 010-041.

If installing a new turbocharger, install the oil supply fitting. Refer to Procedure 010-046.



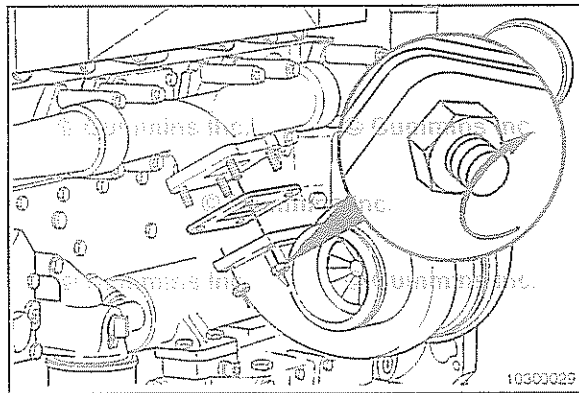


Wastegated

NOTE: For wastegated turbochargers, refer to Procedure 010-050 for wastegate actuator installation.



Apply a film of anti-seize compound, Part Number 3824879, or equivalent, to the turbocharger mounting studs.



Use a new gasket when installing the turbocharger.

Install and tighten the four mounting nuts.



Torque Value: 61 N•m [45 ft-lb]

Air Starting Motor

General Information

The air starting motor system (tanks, line sizes, and valves) is designed and installed by the OEM and starting motor suppliers. Refer any questions about the air starting systems to the manufacturer.

Air Starting Motor Maintenance

- Do not operate the air starting motor with air pressure lower than 480 kPa [70 psi].
- Maintain the air compressor according to the recommendations outlined in the manual.
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- Refer to the OEM and starting motor manufacturer's manuals for specific information regarding the starting motors, valves, and systems.

Refrigerant Compressor Drive

General Information

Industrial Applications

A new refrigerant compressor drive system is offered on the QSM11 industrial. The new drive system has an accessory drive pulley, a six-rib poly-vee belt with automatic tensioning and a backside idler for use with a Sanden rotary compressor.

Remove

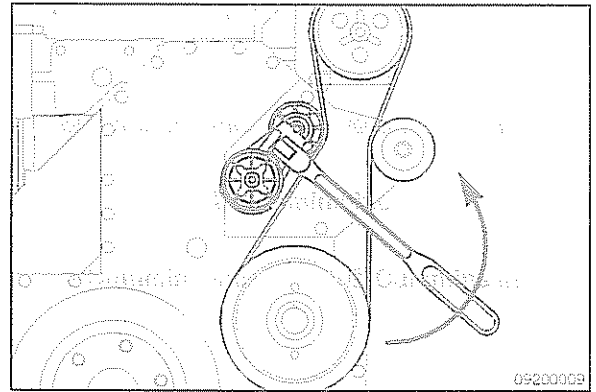
Industrial Applications

Use a 3/8-inch drive breaker bar to hold back the tensioner and relieve the belt tension.

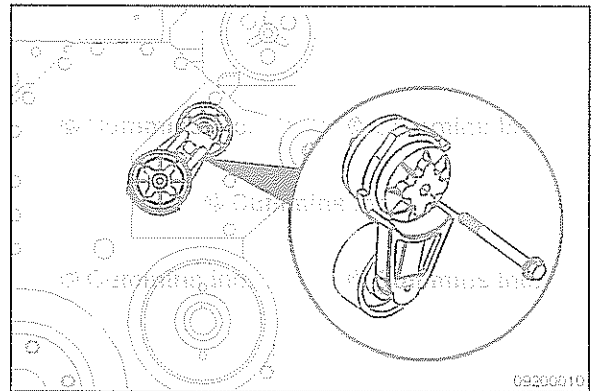
Rotate the tensioner away from the belt.

Remove the refrigerant drive belt while holding the tensioner back.

Relieve the tensioner to neutral position.



Remove the belt tensioner by removing the nut located on top of the tensioner.



Install

Industrial Applications

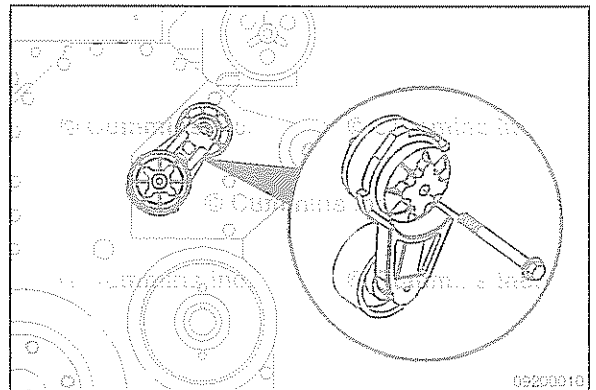
Position the enclosed subassembly (idler pulley) (1) and tensioner mounting bracket (2) on the gear cover and mount with the four capscrews (3) and washers.

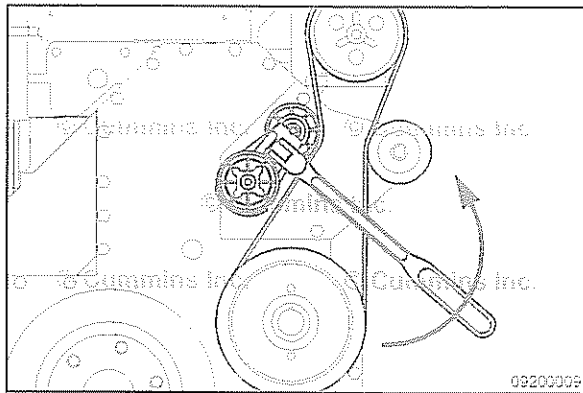
- The tensioner idler pulley will be located in front of two of the mounting capscrew locations in the freestate. The tensioner either will have to be removed completely for this process or the arm moved to one side to have access to all four capscrew mounting locations.
- Install the belt tensioner (4), and tighten the mounting nut.

Torque Value: 47 N•m [35 ft-lb]

Tighten the four mounting capscrews.

Torque Value: 20 N•m [177 in-lb]

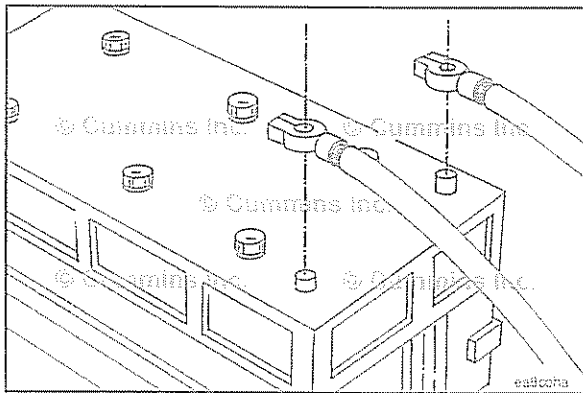




Use a 3/8-inch drive breaker bar to hold back the tensioner.

Install the drive belt while holding the tensioner back.

Allow the tensioner to rotate against the belt.



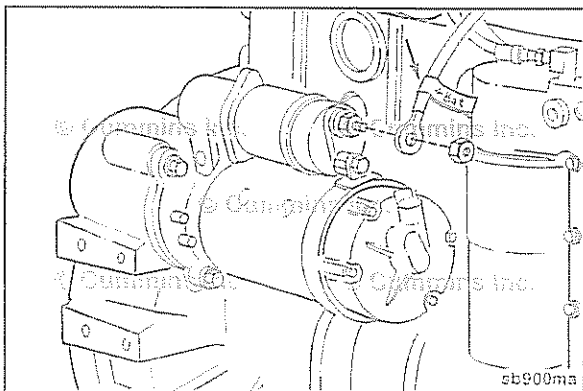
Starting Motor Remove

⚠ WARNING ⚠

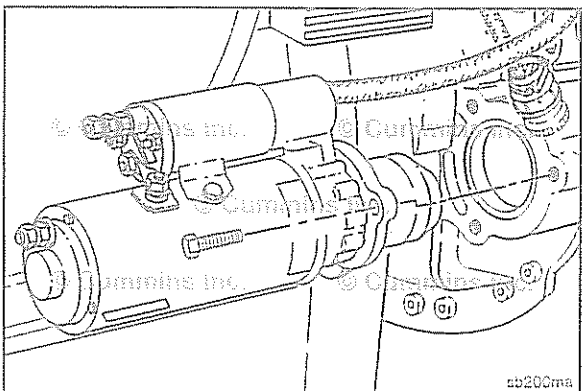
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

NOTE: The starting motor mounting capscrews can be metric or standard thread sizes. Make sure to install the same-size capscrews that were removed.

Remove the electrical connections from the batteries.



Tag and remove the electrical connections from the starting motor.



Remove the three capscrews and the starting motor.

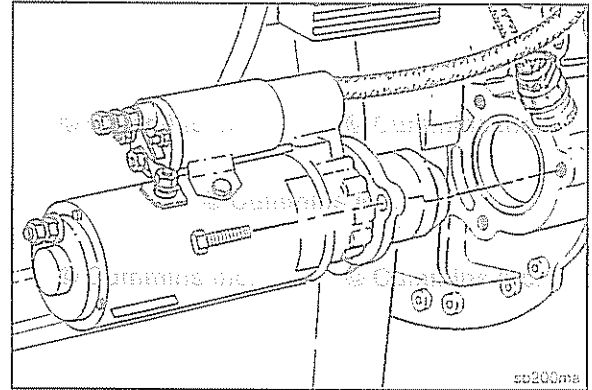
Install

Before installing the starting motor, make sure the capscrews are the correct size and grade.

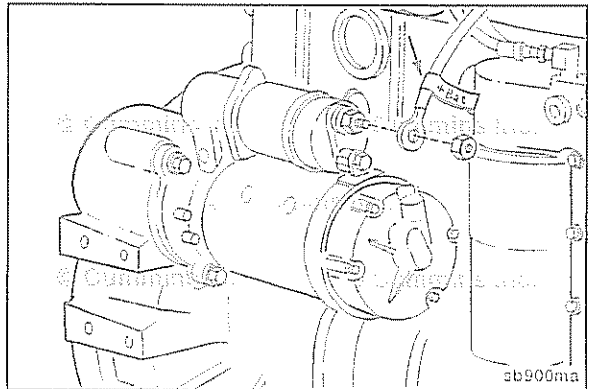
Install the starting motor and the three mounting capscrews.

Tighten the capscrews.

Torque Value: 90 N•m [66 ft-lb]



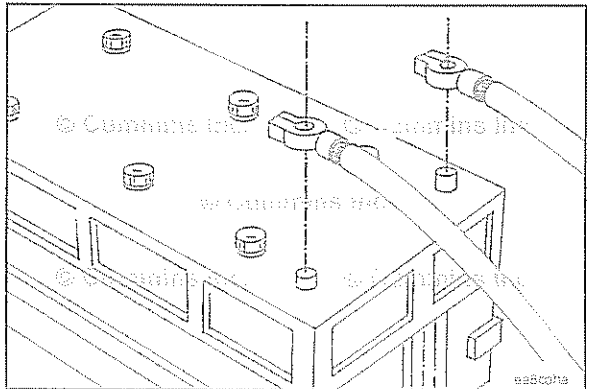
Install and tighten the electrical connections to the starting motor.



WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

Install and tighten the battery electrical connections.



Engine Storage - Long Term

General Information

If the engine will be out of service longer than 6 months, take special precautions to prevent rust. Contact the nearest Cummins Authorized Repair Location for information concerning engine storage procedures.